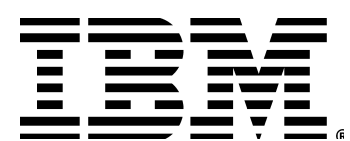


Containers & OpenShift Workshop

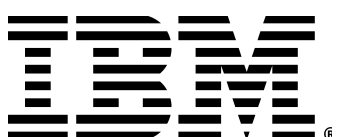
Alexander Mamani - IBM
Kate Queiroz - Red Hat
Christian Zaccaria - Red Hat
Winnie Imafidon - IBM



Let's do a recap of kubernetes



Red Hat



1. RECAP

a container is the smallest compute unit



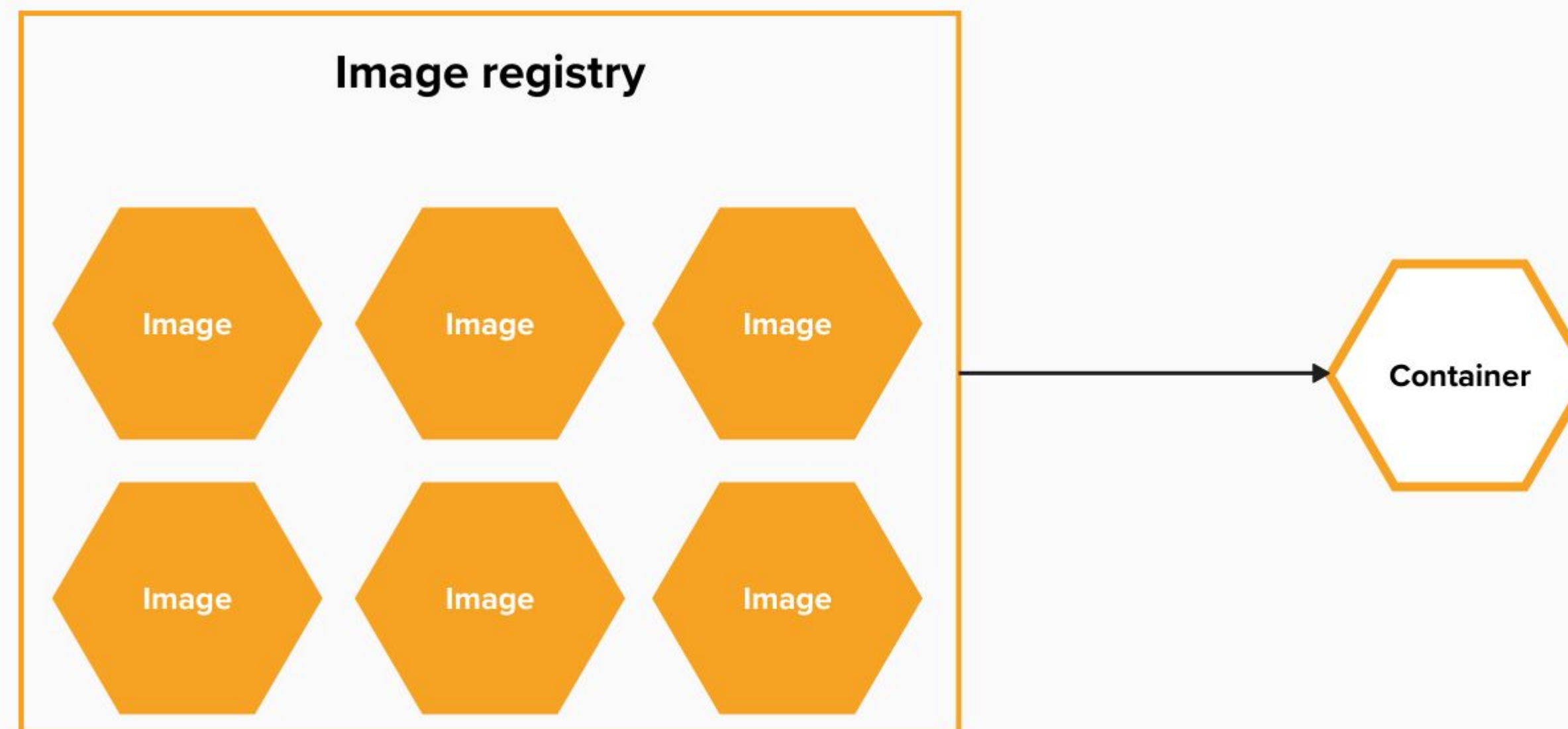
1. RECAP

containers are created from container images



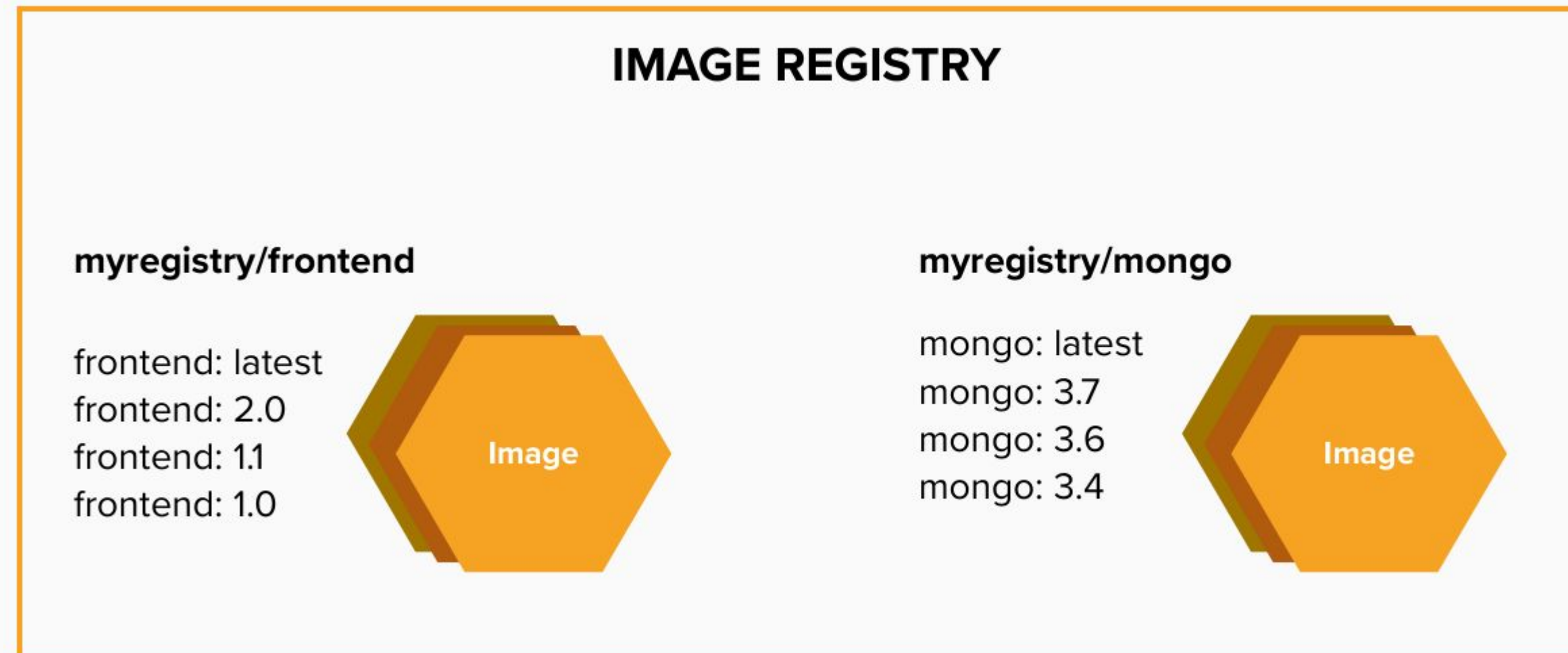
1. RECAP

container images are stored in an image registry



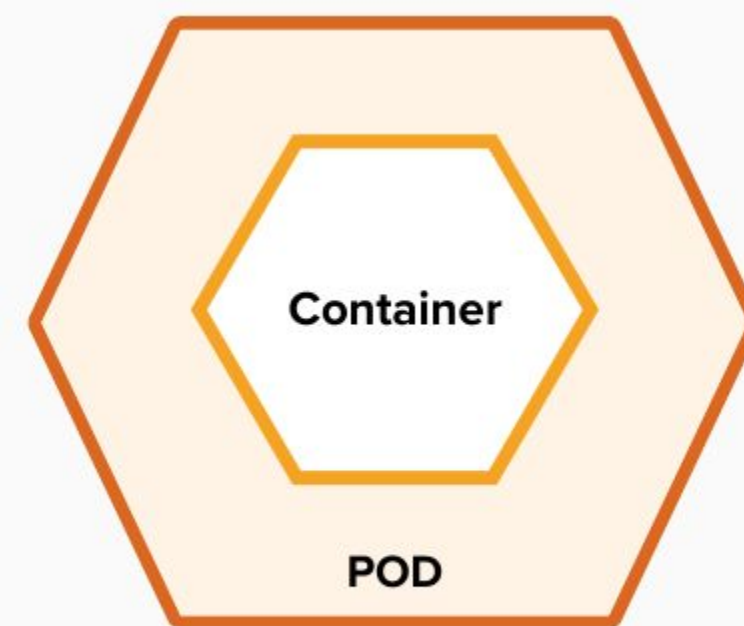
1. RECAP

an image repository contains all versions of an image in the image registry

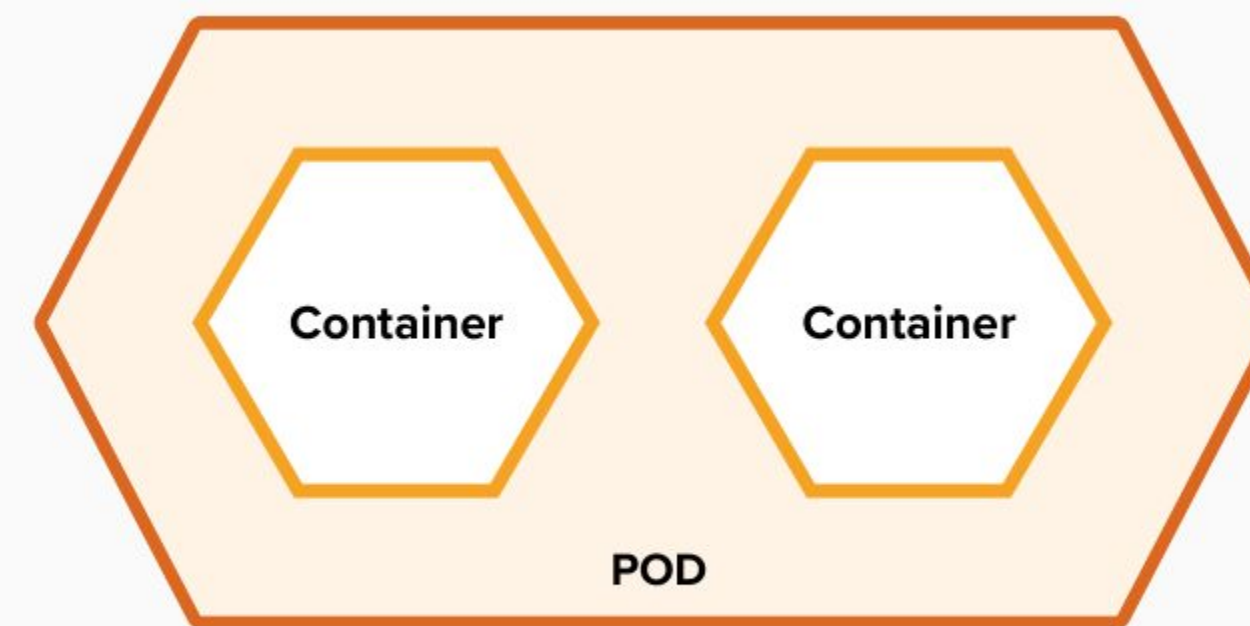


1. RECAP

containers are wrapped in pods which are units of deployment and management



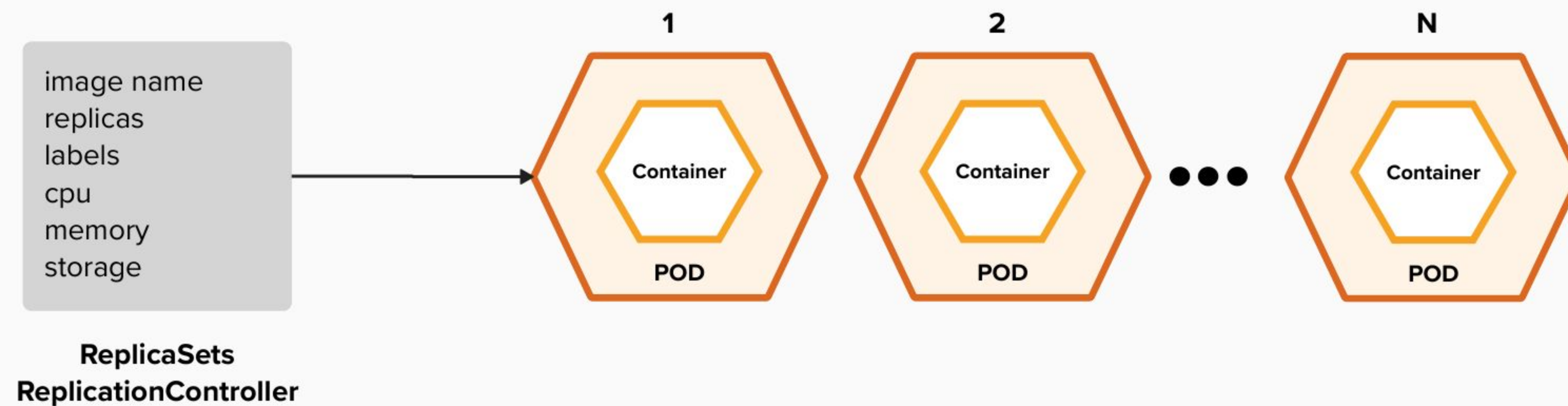
10.140.4.44



10.15.6.55

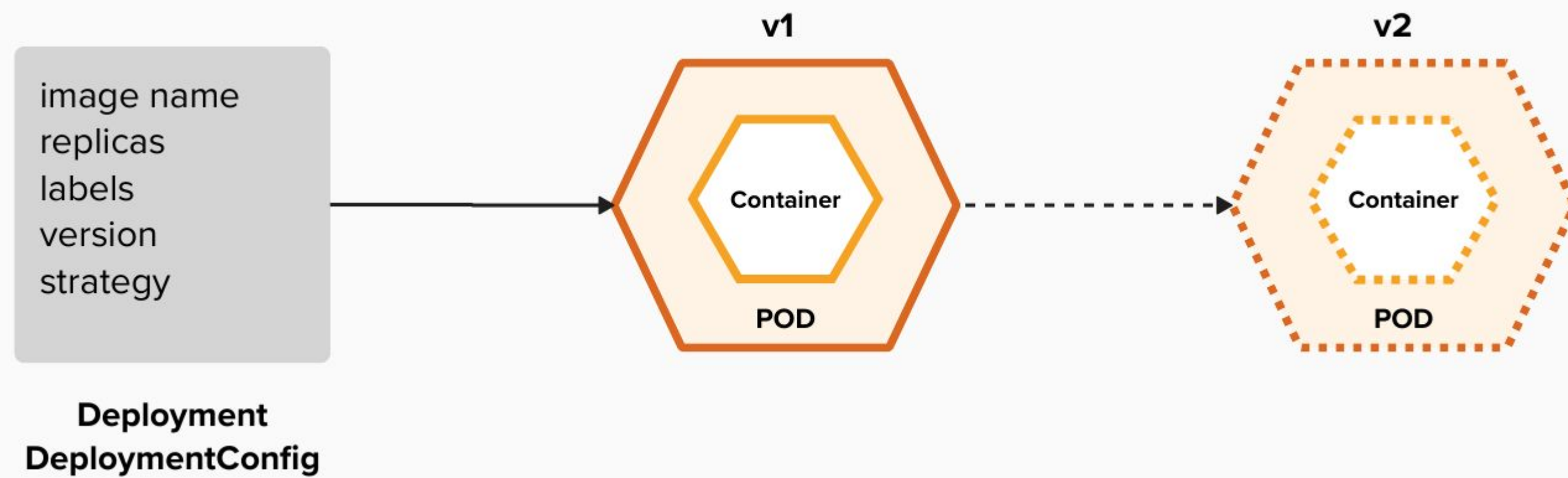
1. RECAP

ReplicationControllers & ReplicaSets ensure a specified number of pods are running at any given time



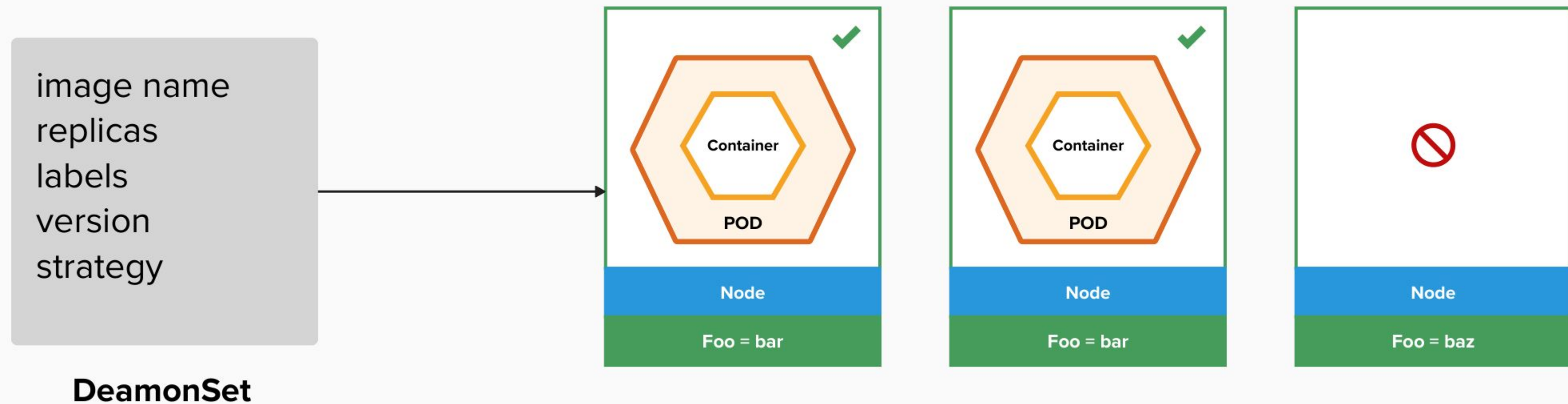
1. RECAP

Deployments and DeploymentConfigurations
define how to roll out new versions of Pods



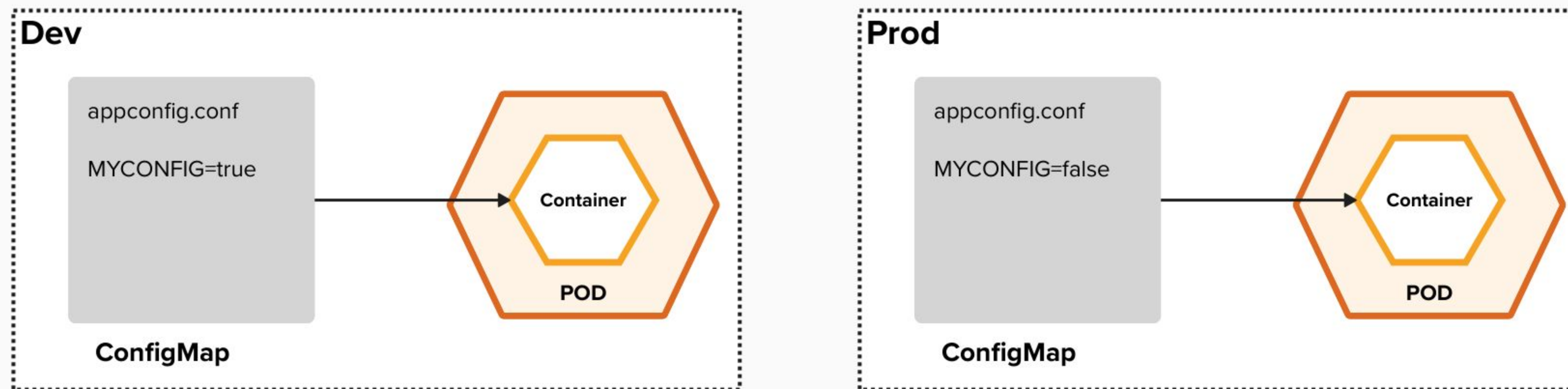
1. RECAP

a daemonset ensures that all (or some) nodes run
a copy of a pod



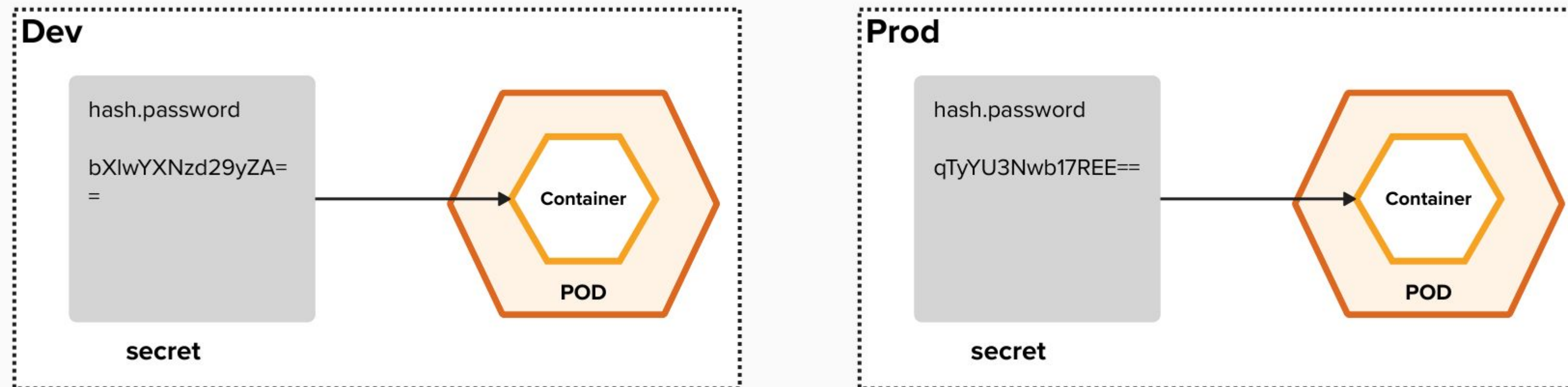
1. RECAP

configmaps allow you to decouple configuration artifacts from image content



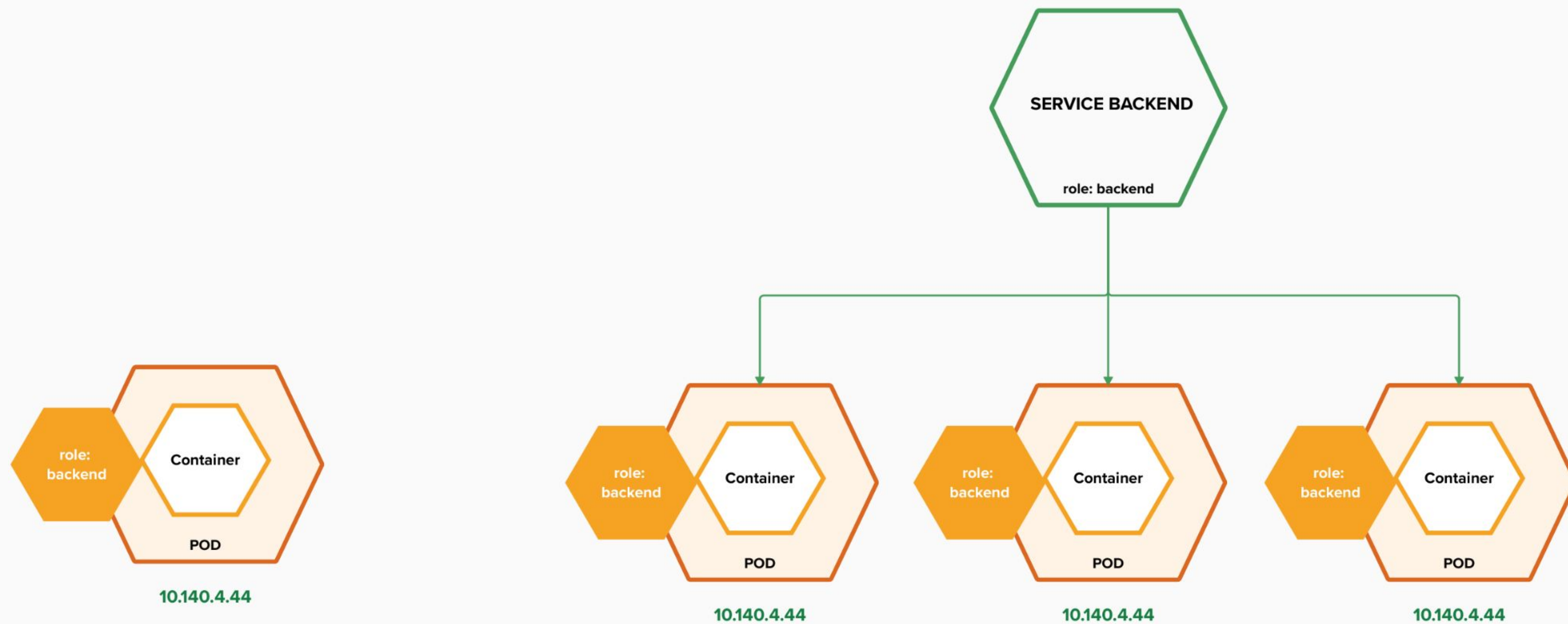
1. RECAP

secrets provide a mechanism to hold sensitive information such as passwords



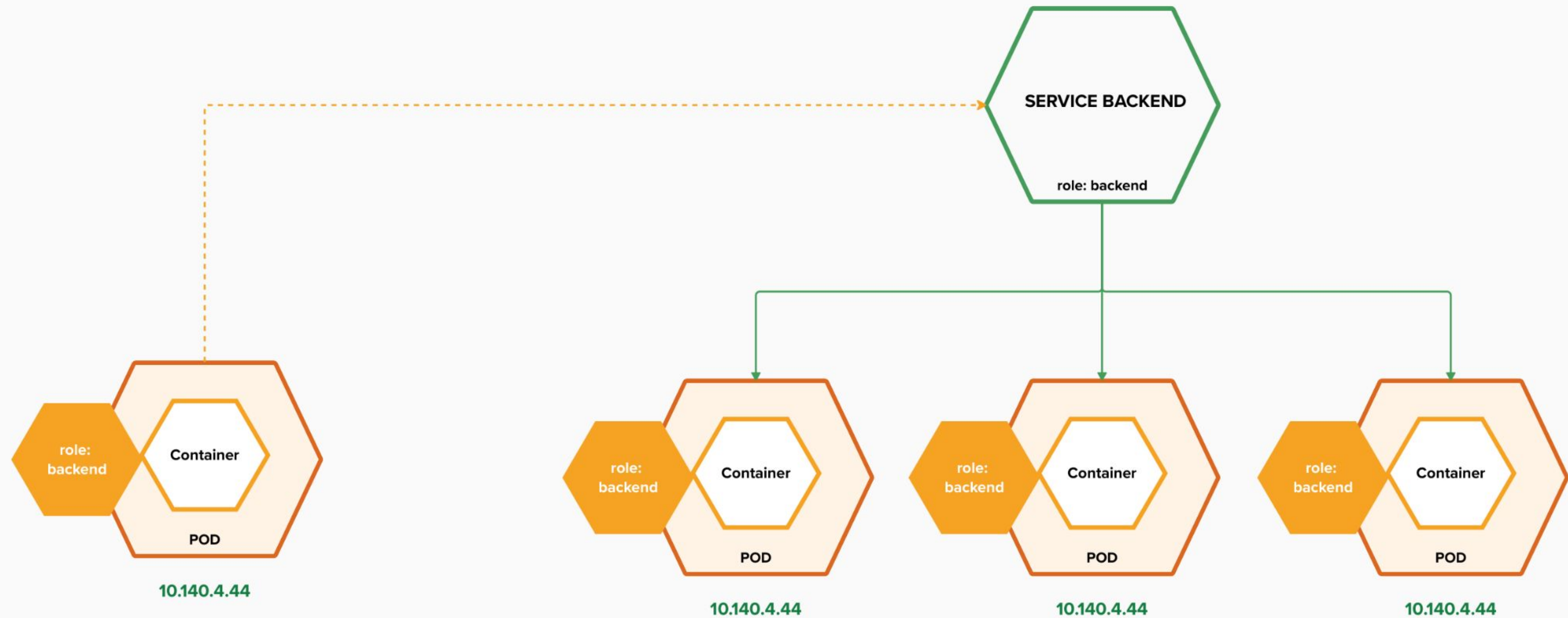
1. RECAP

services provide internal load-balancing and
service discovery across pods



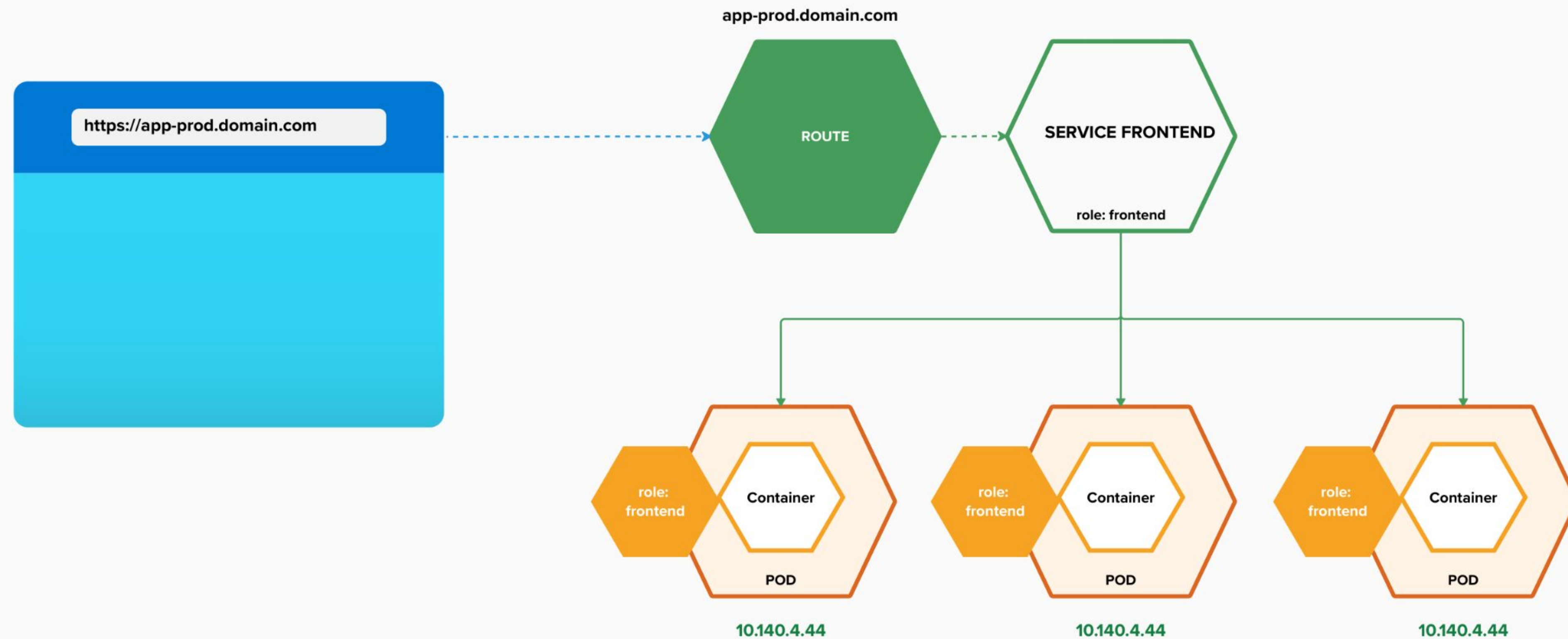
1. RECAP

apps can talk to each other via services



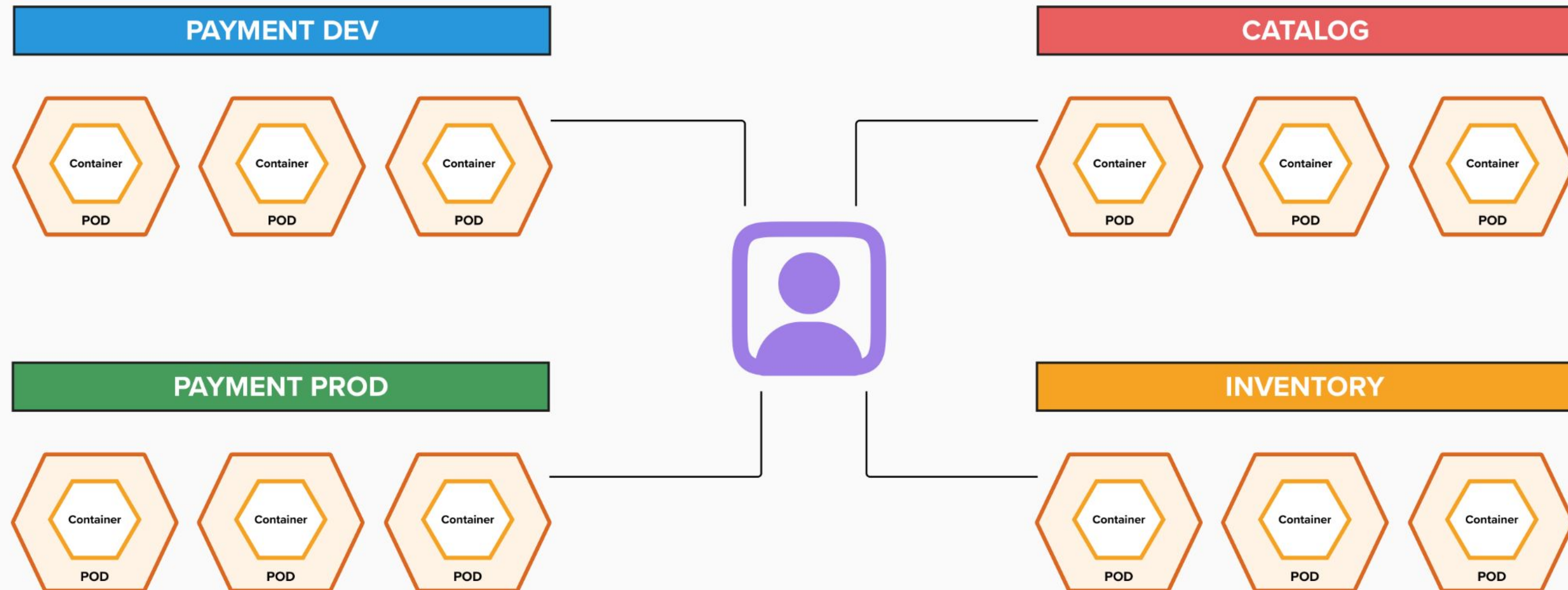
1. RECAP

routes make services accessible to clients outside the environment via real-world urls



1. RECAP

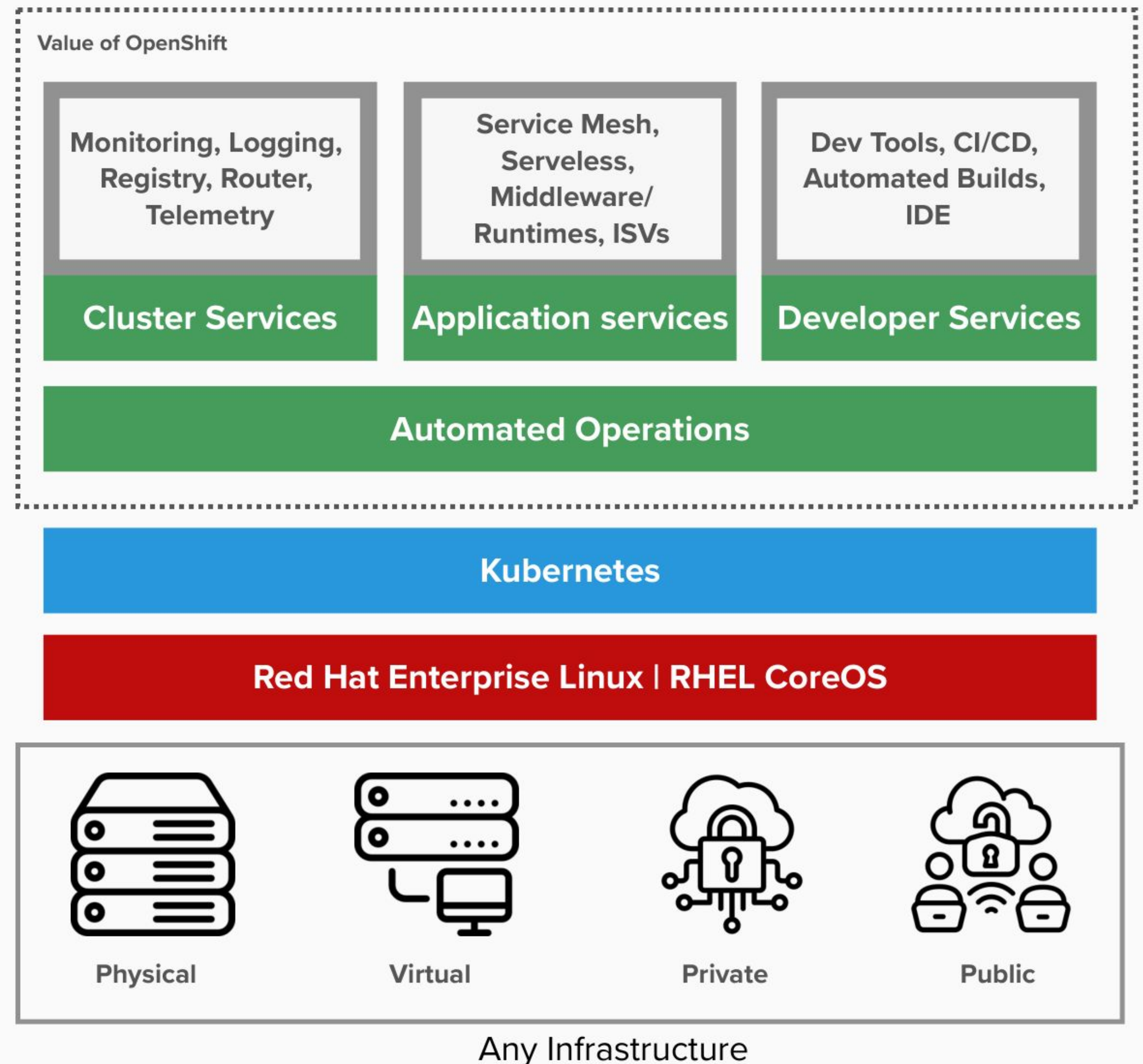
projects isolate apps across environments, teams,
groups and departments



What is OpenShift?

Everything you need, out of the box

1. Fully integrated and automated architecture
2. Seamless Kubernetes deployment on any cloud or on-premises environment
3. Fully automated installation, from cloud infrastructure to OS to application services
4. One click platform and application updates
5. Auto-scaling of cloud resources



OPENSIFT ARCHITECTURE

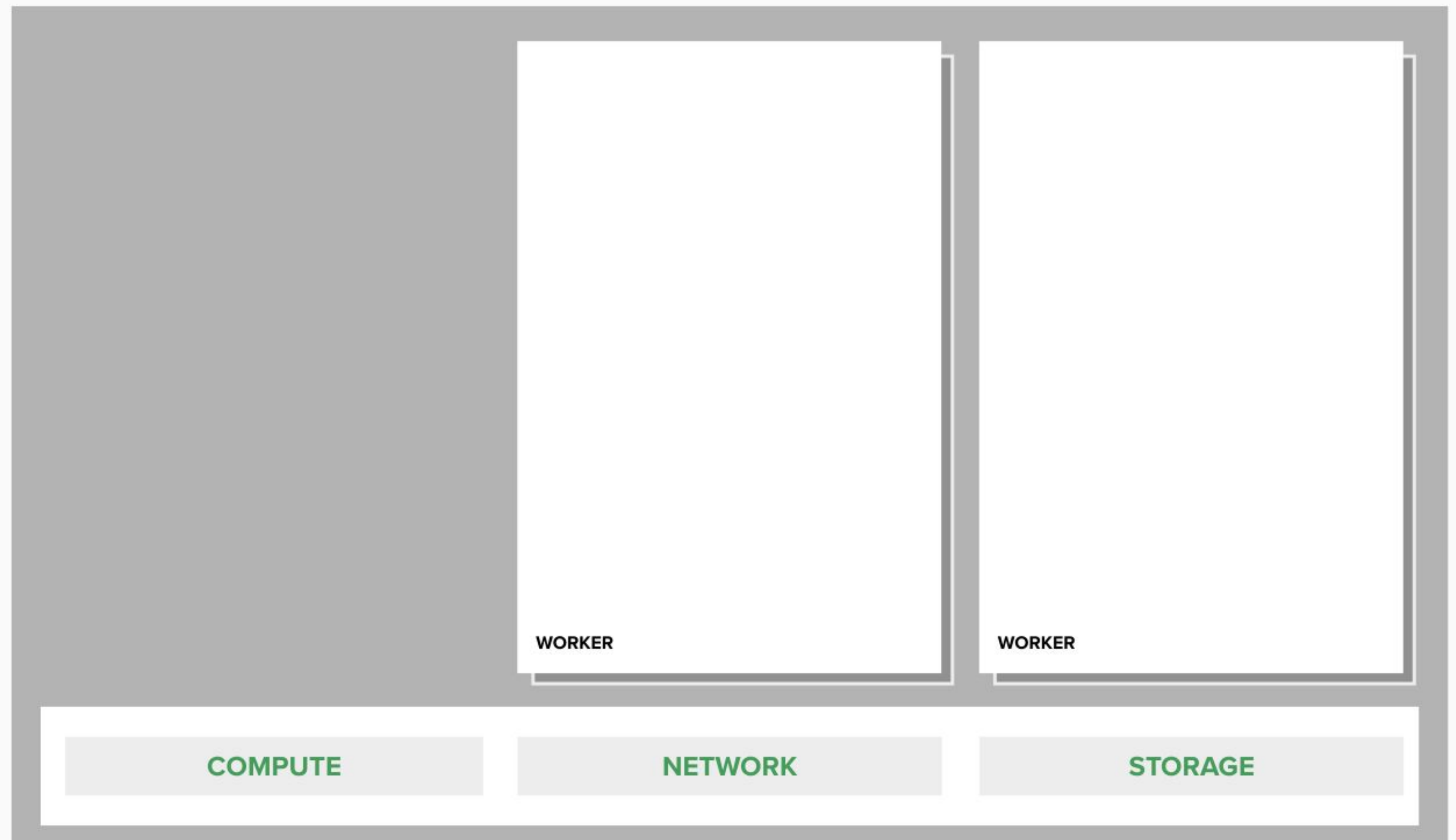
2. OPENSIFT ARCHITECTURE

your choice of infrastructure



2. OPENSIFT ARCHITECTURE

workers run workloads



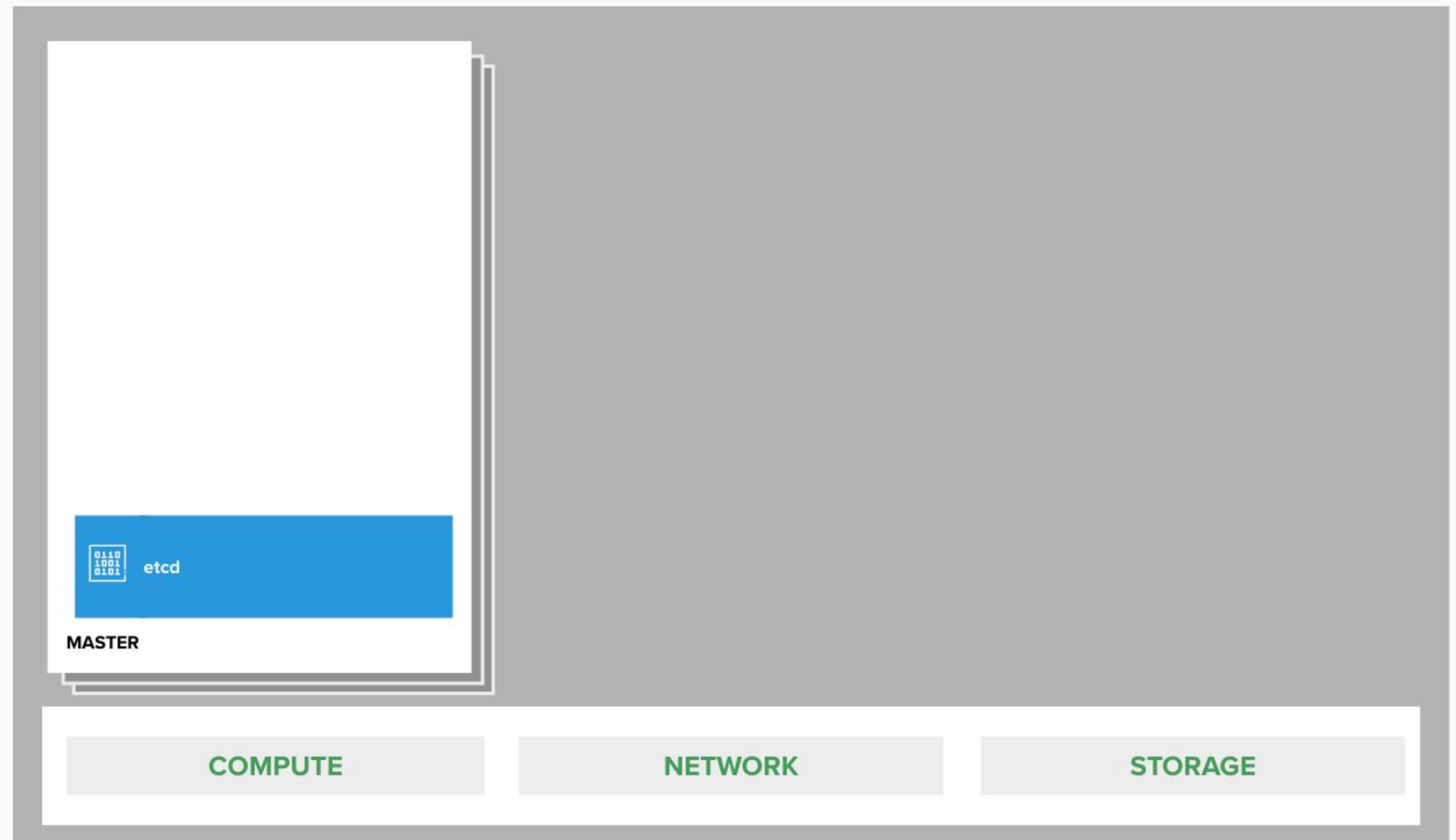
2. OPENSIFT ARCHITECTURE

masters are the control plane



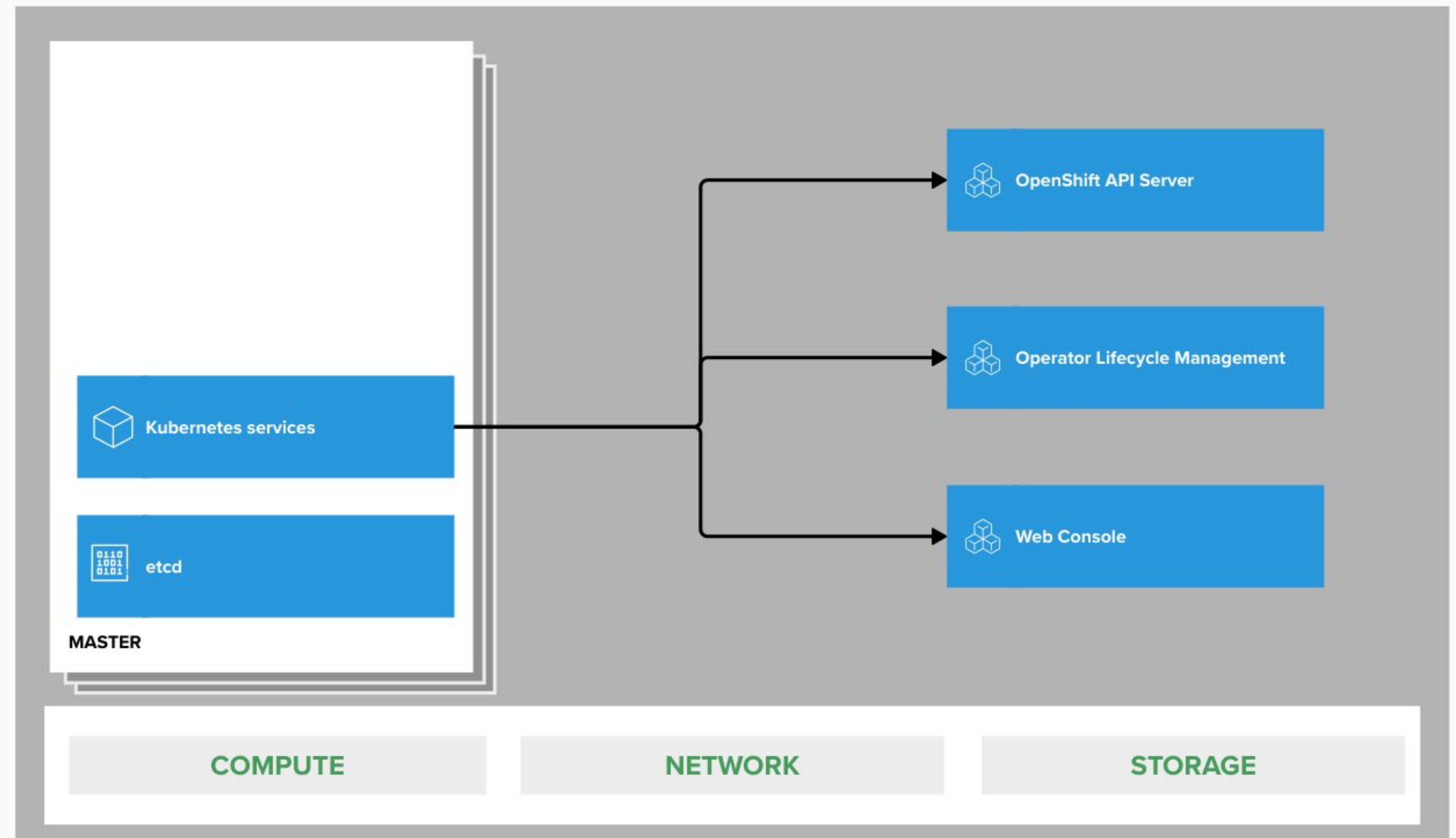
2. OPENSIFT ARCHITECTURE

state of everything



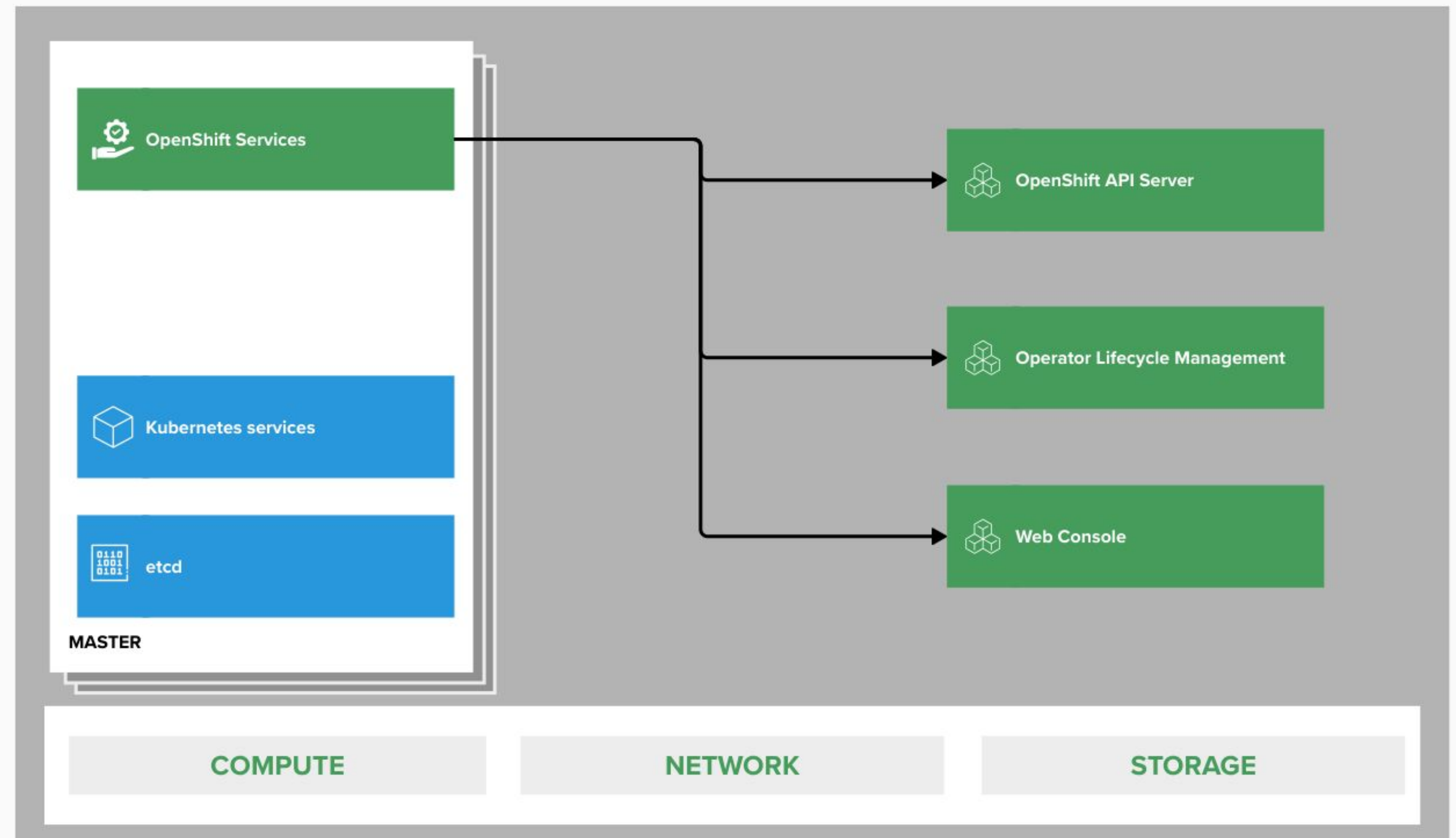
2. OPENSIFT ARCHITECTURE

core kubernetes components



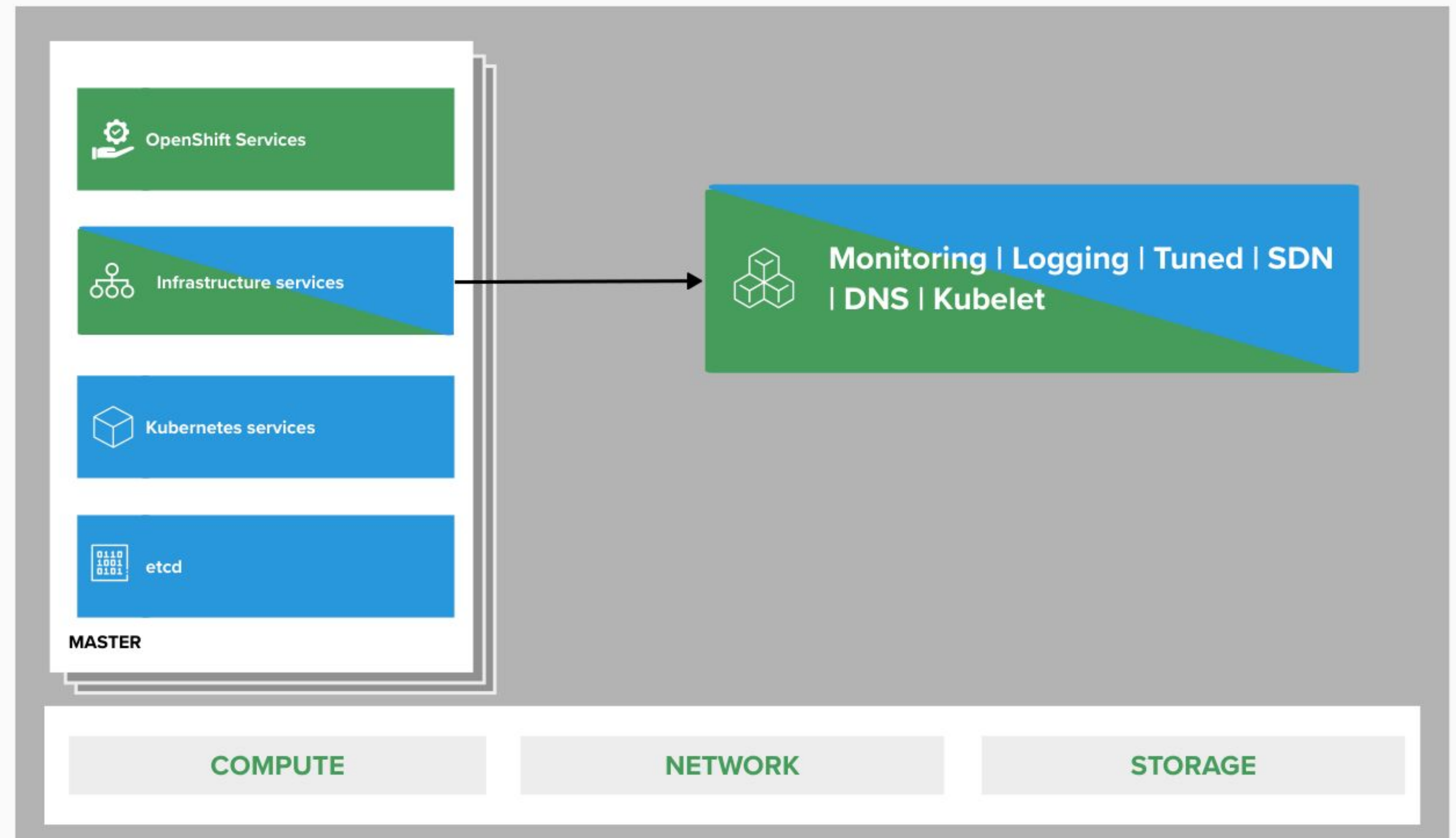
2. OPENSIFT ARCHITECTURE

core OpenShift components



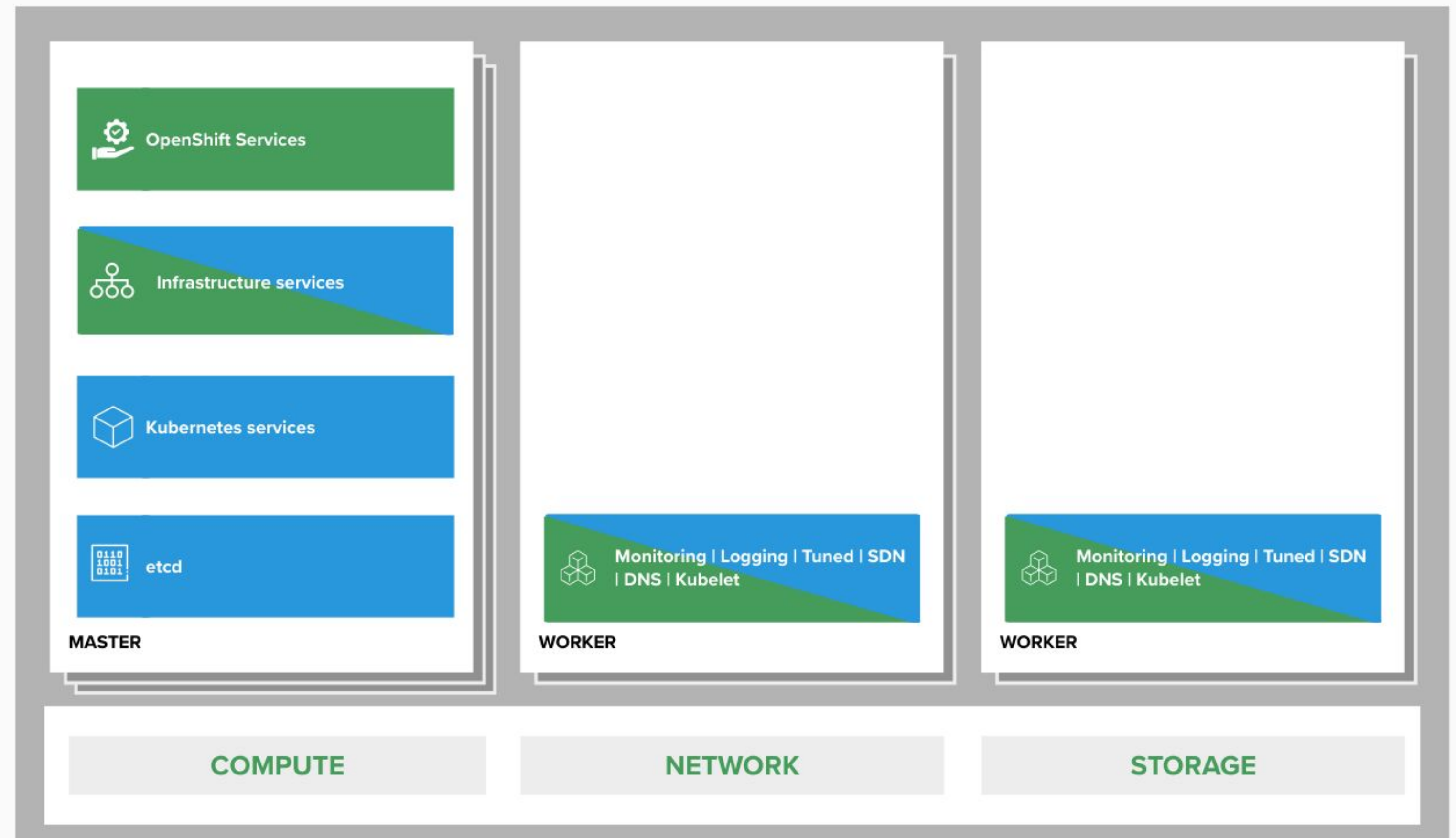
2. OPENSIFT ARCHITECTURE

internal and support infrastructure services



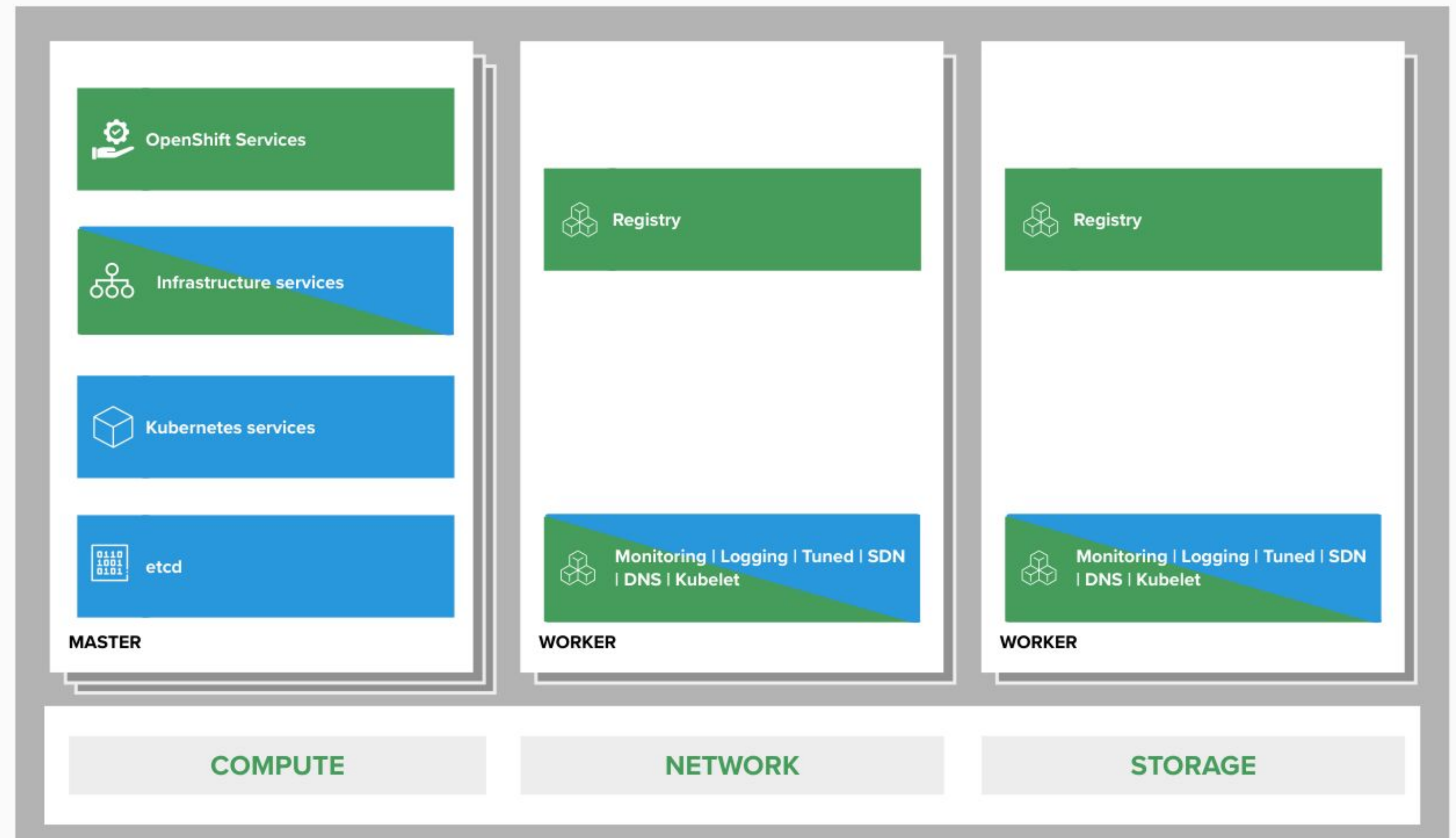
2. OPENSIFT ARCHITECTURE

run on all hosts



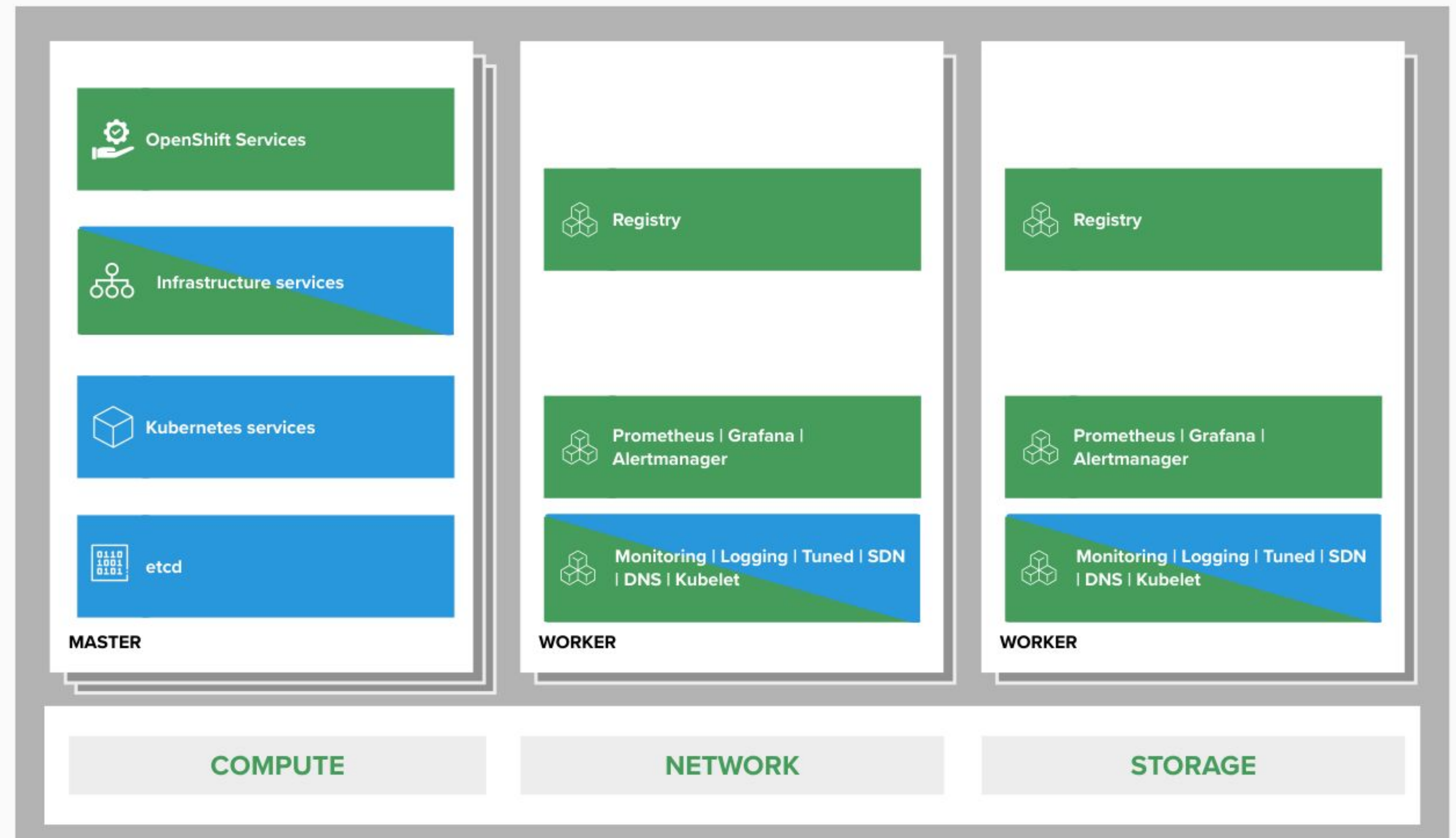
2. OPENSIFT ARCHITECTURE

integrated image registry



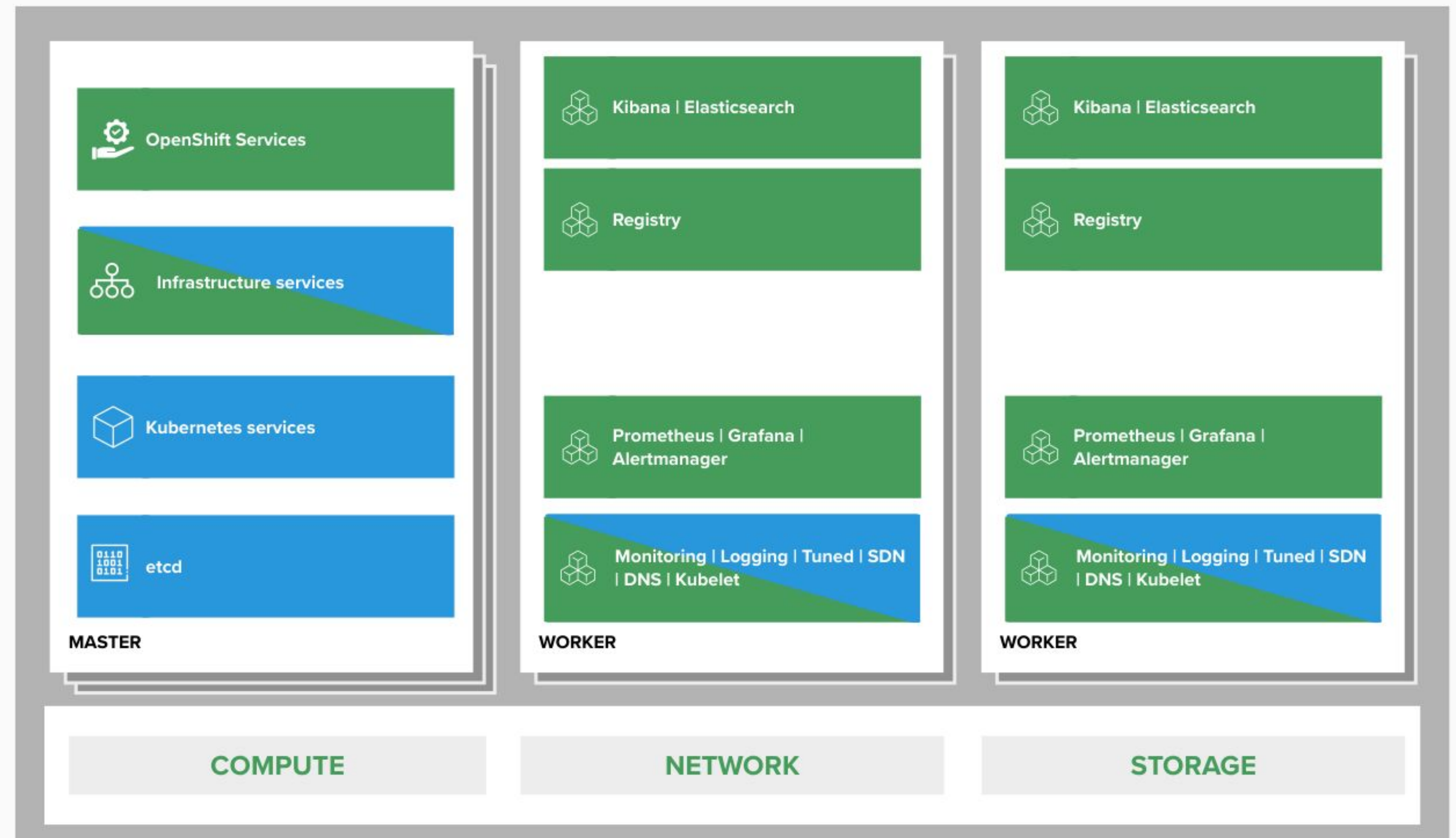
2. OPENSIFT ARCHITECTURE

cluster monitoring



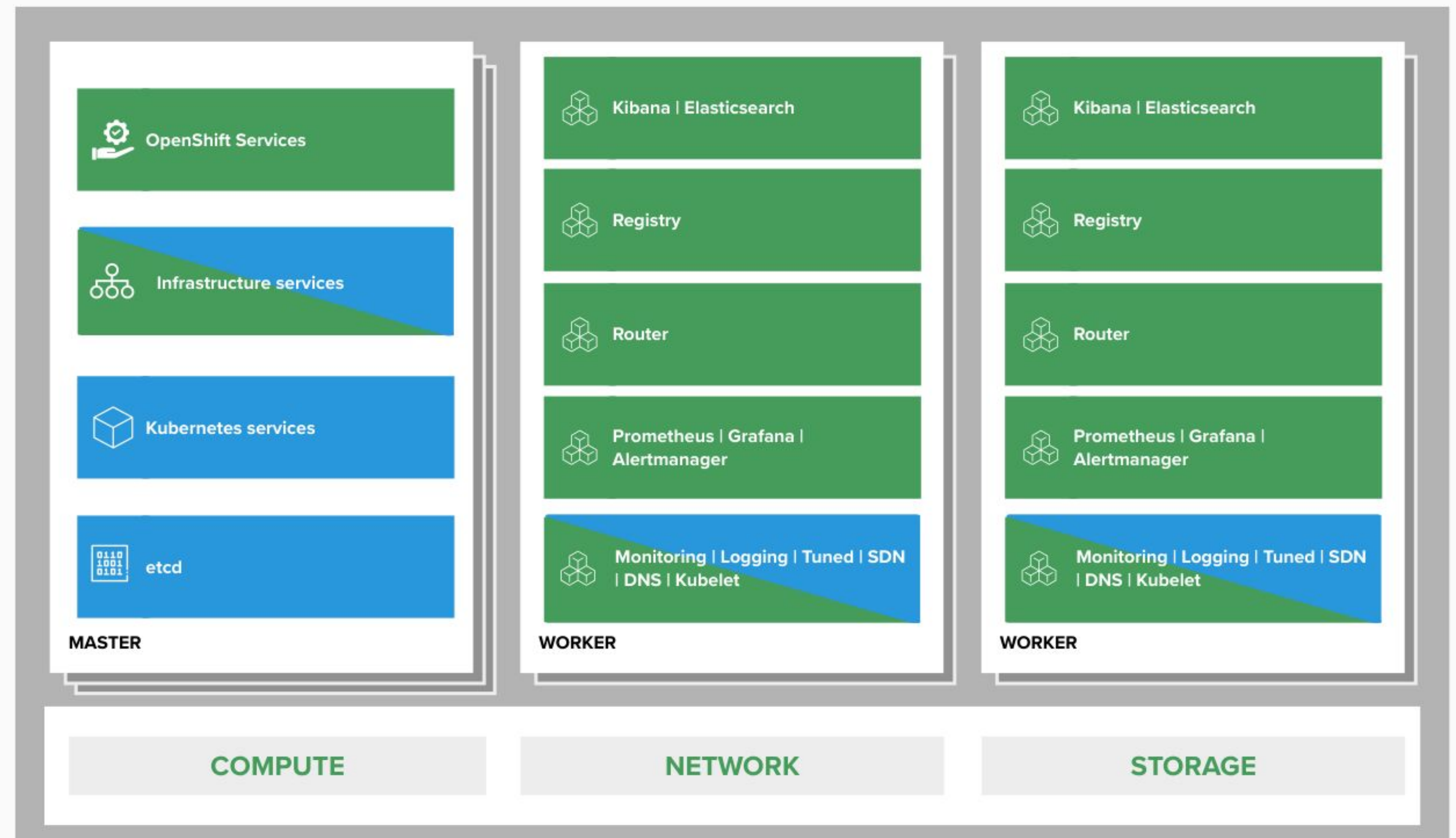
2. OPENSIFT ARCHITECTURE

log aggregation



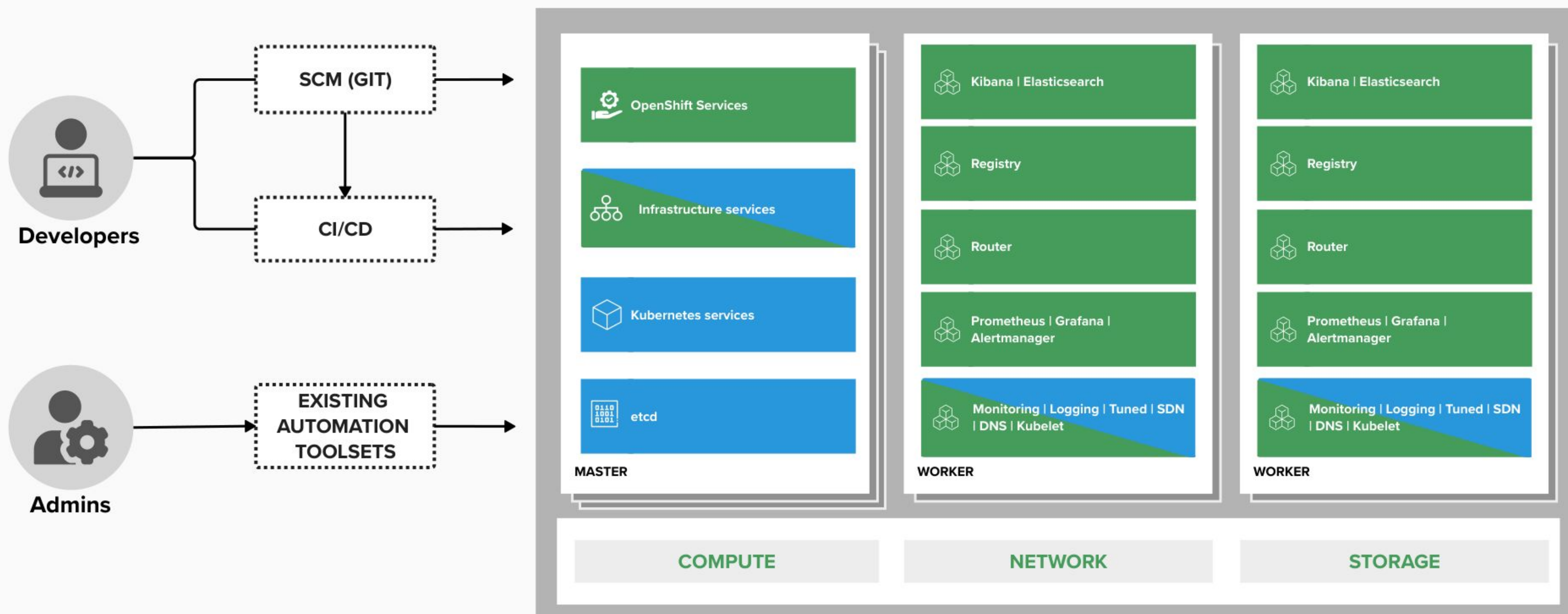
2. OPENSIFT ARCHITECTURE

integrated routing



2. OPENSIFT ARCHITECTURE

dev and ops via web, cli, API, and IDE



OpenShift lifecycle, installation & upgrades



Red Hat



Red Hat Enterprise Linux CoreOS

	RED HAT® ENTERPRISE LINUX® General Purpose OS	RED HAT® ENTERPRISE LINUX CoreOS Immutable container host
BENEFITS	• 10+ year enterprise life cycle • Industry standard security • High performance on any infrastructure • Customizable and compatible with wide ecosystem of partner solutions	• Self-managing, over-the-air updates • Immutable and tightly integrated with OpenShift • Host isolation is enforced via Containers • Optimized performance on popular infrastructure
WHEN TO USE	When customization and integration with additional solutions is required	When cloud-native, hands-free operations are a top priority

Immutable Operating System

Red Hat Enterprise Linux CoreOS is versioned with OpenShift

CoreOS is tested and shipped in conjunction with the platform. Red Hat runs thousands of tests against these configurations.

Red Hat Enterprise Linux CoreOS is managed by the cluster

The Operating system is operated as part of the cluster, with the config for components managed by Machine Config Operator:

- CRI-O config
- Kubelet config
- Authorized registries
- SSH config

RHEL CoreOS admins are responsible for:

Nothing.



Red Hat
OpenShift
Container Platform

v4.22.3



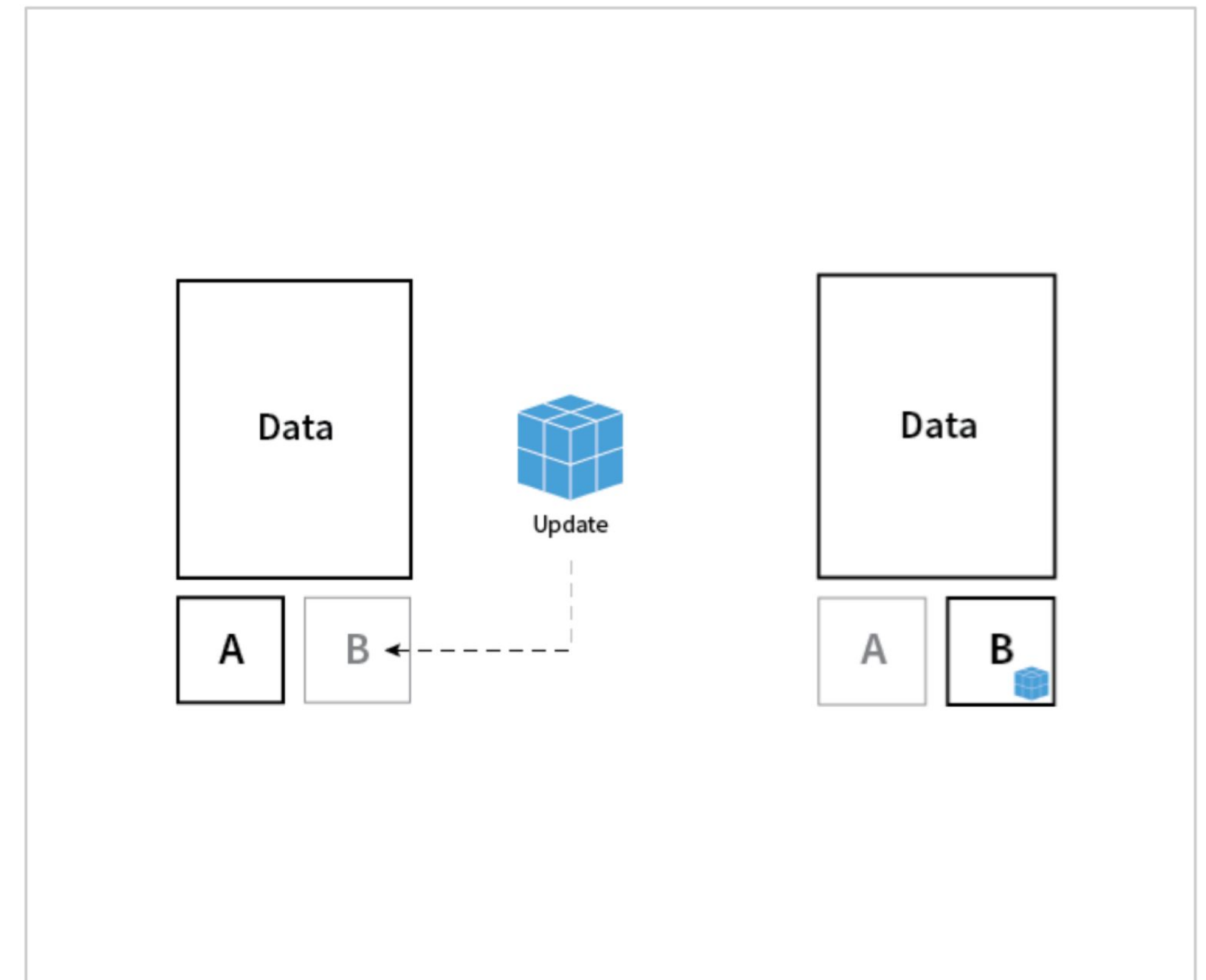
Red Hat
Enterprise Linux
CoreOS

v4.22.3

Transactional Updates via rpm-ostree

Transactional updates ensure that the Red Hat CoreOS is never altered during runtime. Rather it is booted directly into an always “known good” version.

- Each OS update is versioned and tested as an complete image.
- OS binaries (/usr) are read-only
- Updates encapsulated in container images
- file system and package layering available for hotfixes and debugging



Machine Config Operator (MCO)

Provides cluster-level configuration, enables rolling upgrades, and prevents drift between new and existing nodes. The MCO is the heart of what makes RHCOS a kube-native operating system.

Configure Kernel Arguments for the Cluster

- `oc create -f 50-kargs.yaml`
- `oc edit mc/50-kargs`

MCO can be paused to suspend operations

Provides control for when changes can be deployed

Custom MachinePools can have inheritance

Enables MachineConfigs to scale

```
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: worker
  name: 50-kargs
spec:
  KernelArguments:
    - audit=1
    - audit_backlog_limit=8192
    - net.ifnames.prefix=net
```


CRI-O Support in OpenShift

CRI-O tracks and versions identical to Kubernetes, simplifying support permutations

CRI-O 1.34



Kubernetes 1.34



OpenShift 4.21



CRI-O 1.35



Kubernetes 1.35



OpenShift 4.22



CRI-O 1.36



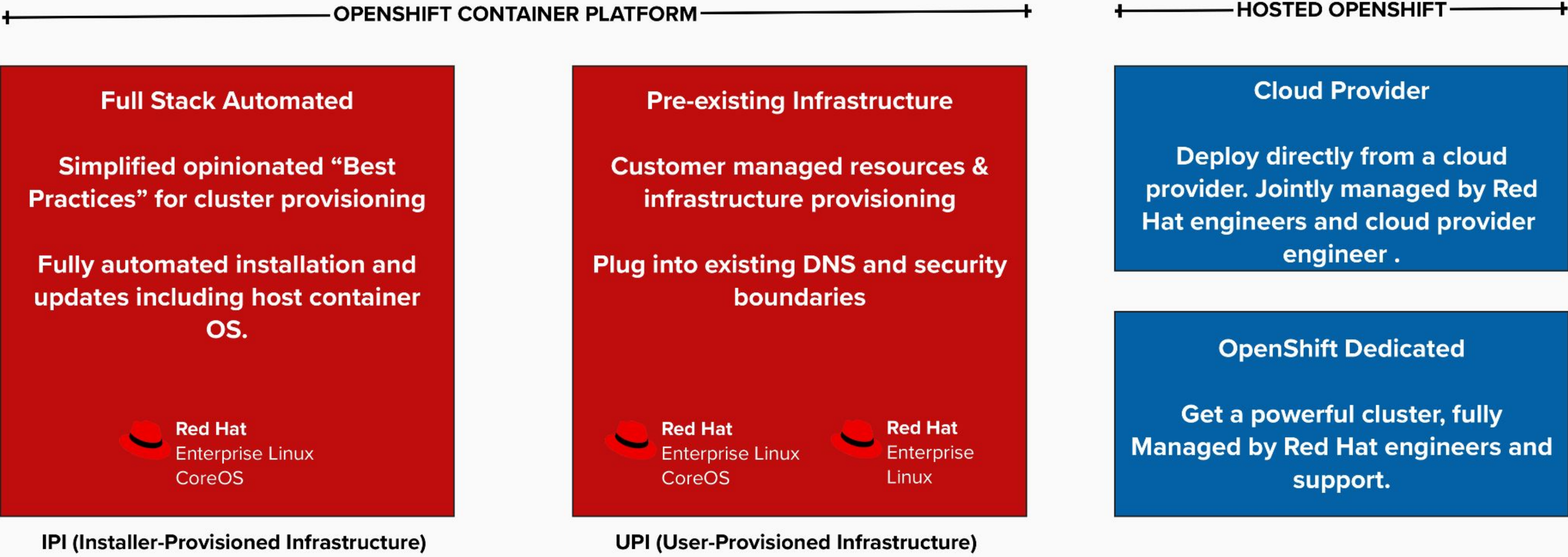
Kubernetes 1.36



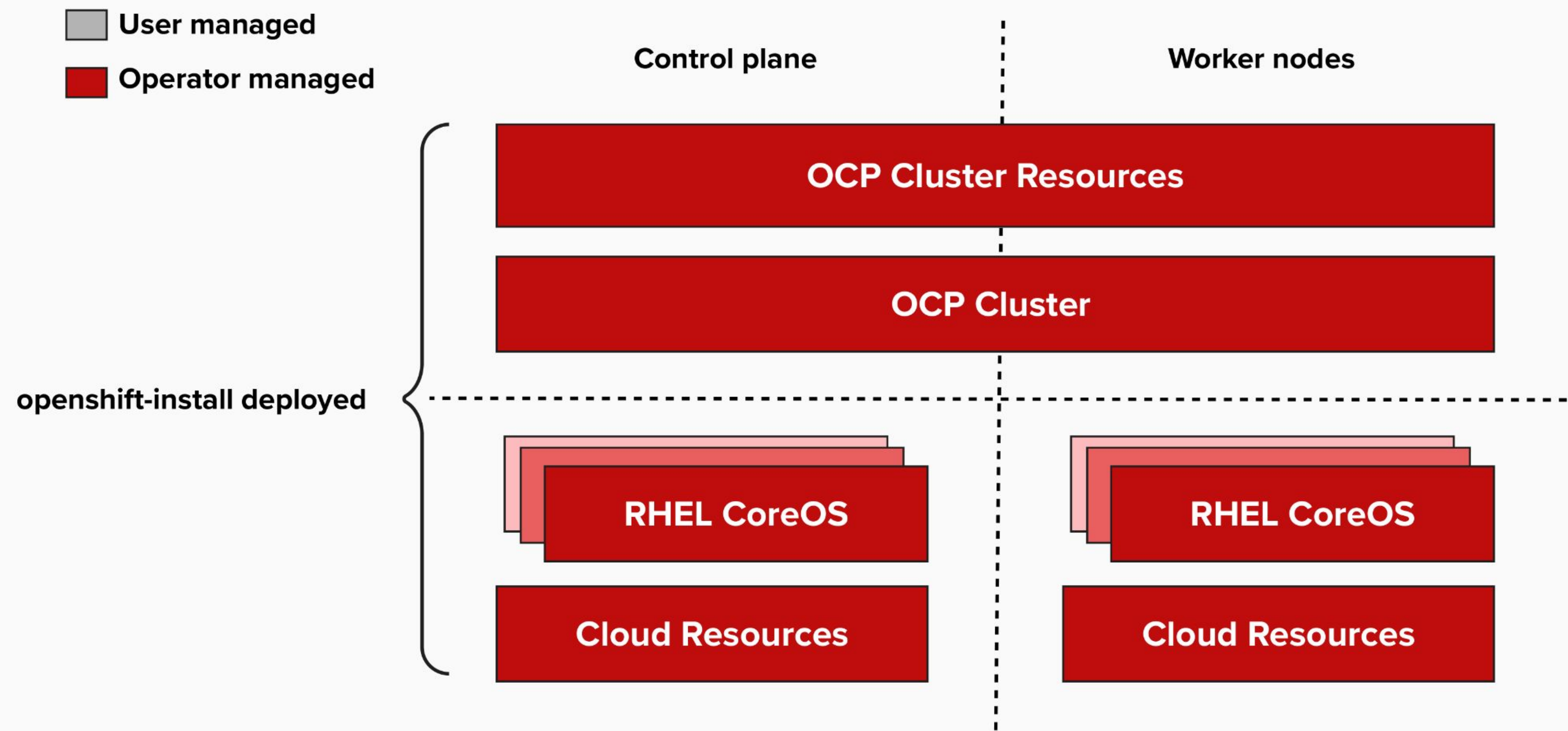
OpenShift 5.0



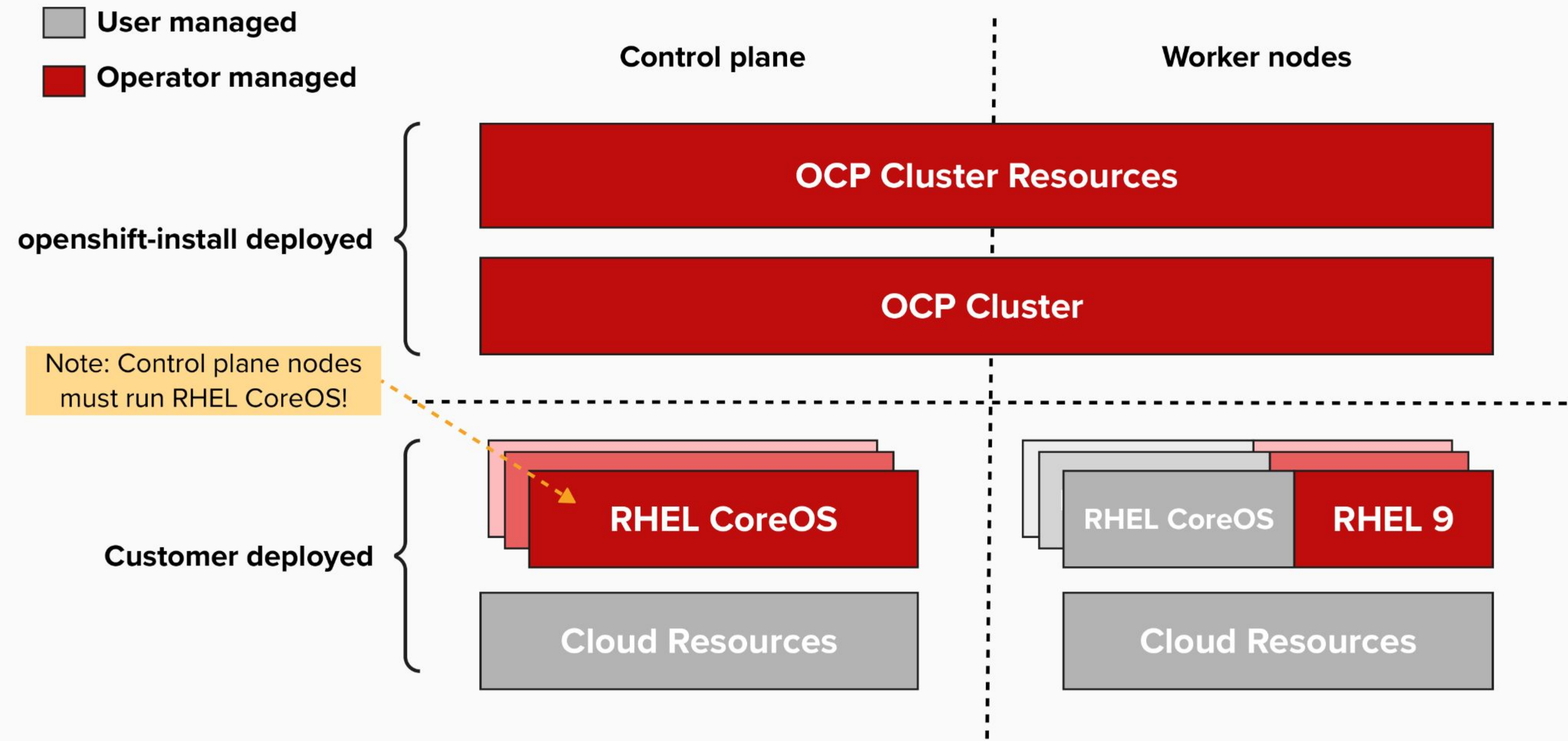
Installation



Full-stack Automated Installation



Pre-existing Infrastructure Installation



Comparison of Paradigms

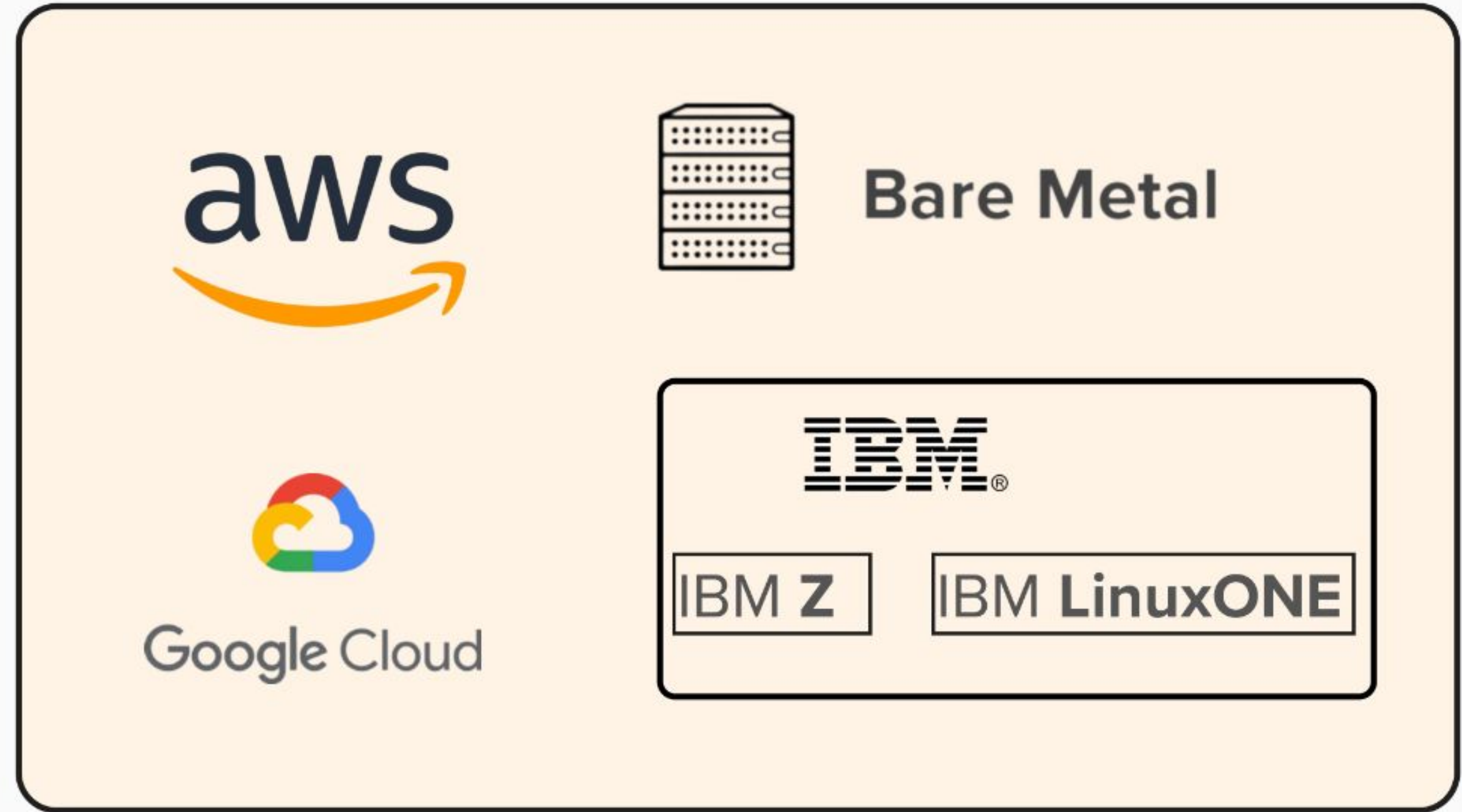
	Full Stack Automation	Pre-existing Infrastructure
Build Network	Installer	User
Setup Load Balancers	Installer	User
Configure DNS	Installer	User
Hardware/VM Provisioning	Installer	User
OS Installation	Installer	User
Generate Ignition Configs	Installer	Installer
OS Support	Installer: RHEL CoreOS	User: RHEL CoreOS + RHEL 9
Node Provisioning / Autoscaling	yes	Only for providers with OpenShift Machine API support

Supported Providers

Full Stack Automation (IPI)

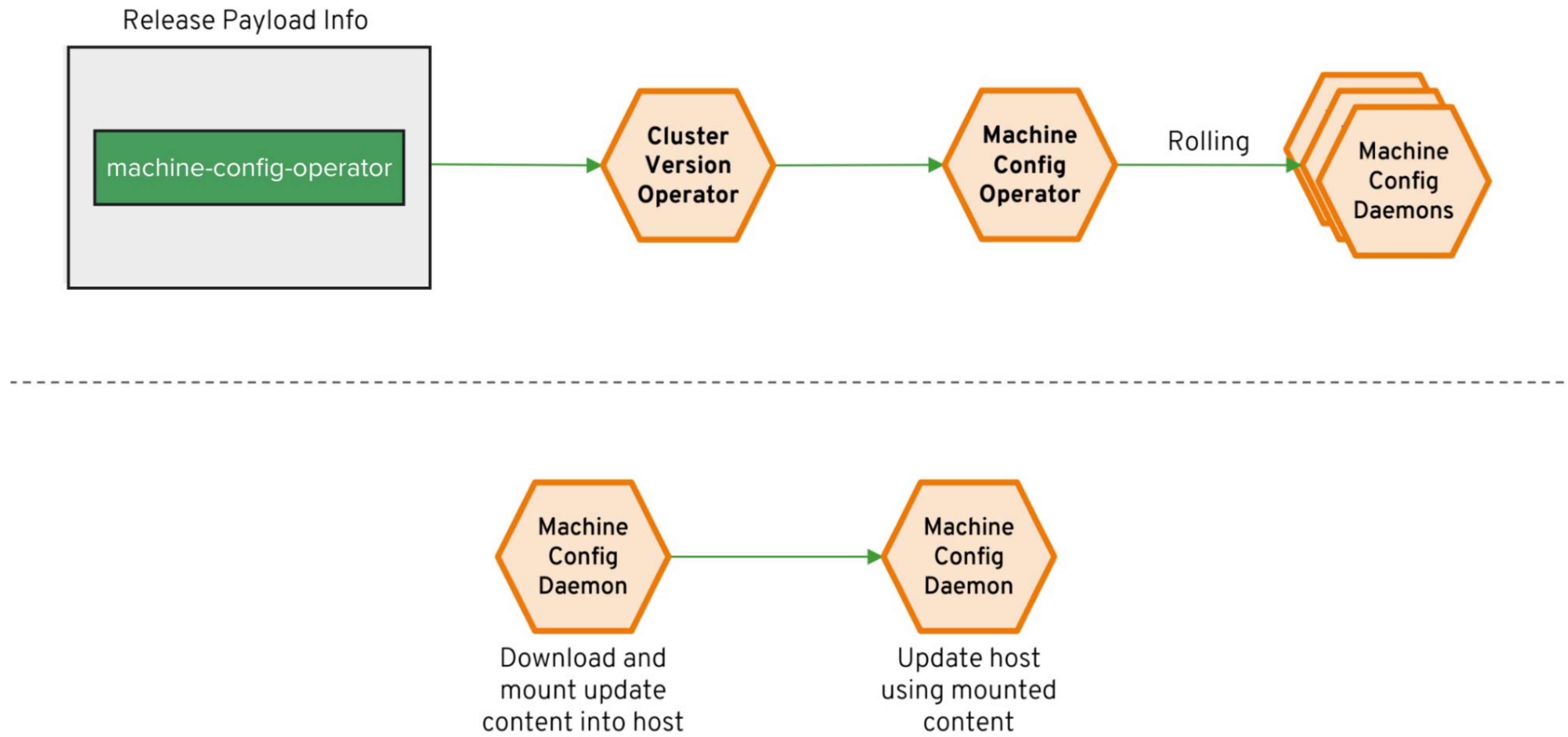


Pre-existing Infrastructure (UPI)



Cluster Management

Over-the-air updates



Cluster Management

Rolling Machine Updates

Single-click updates

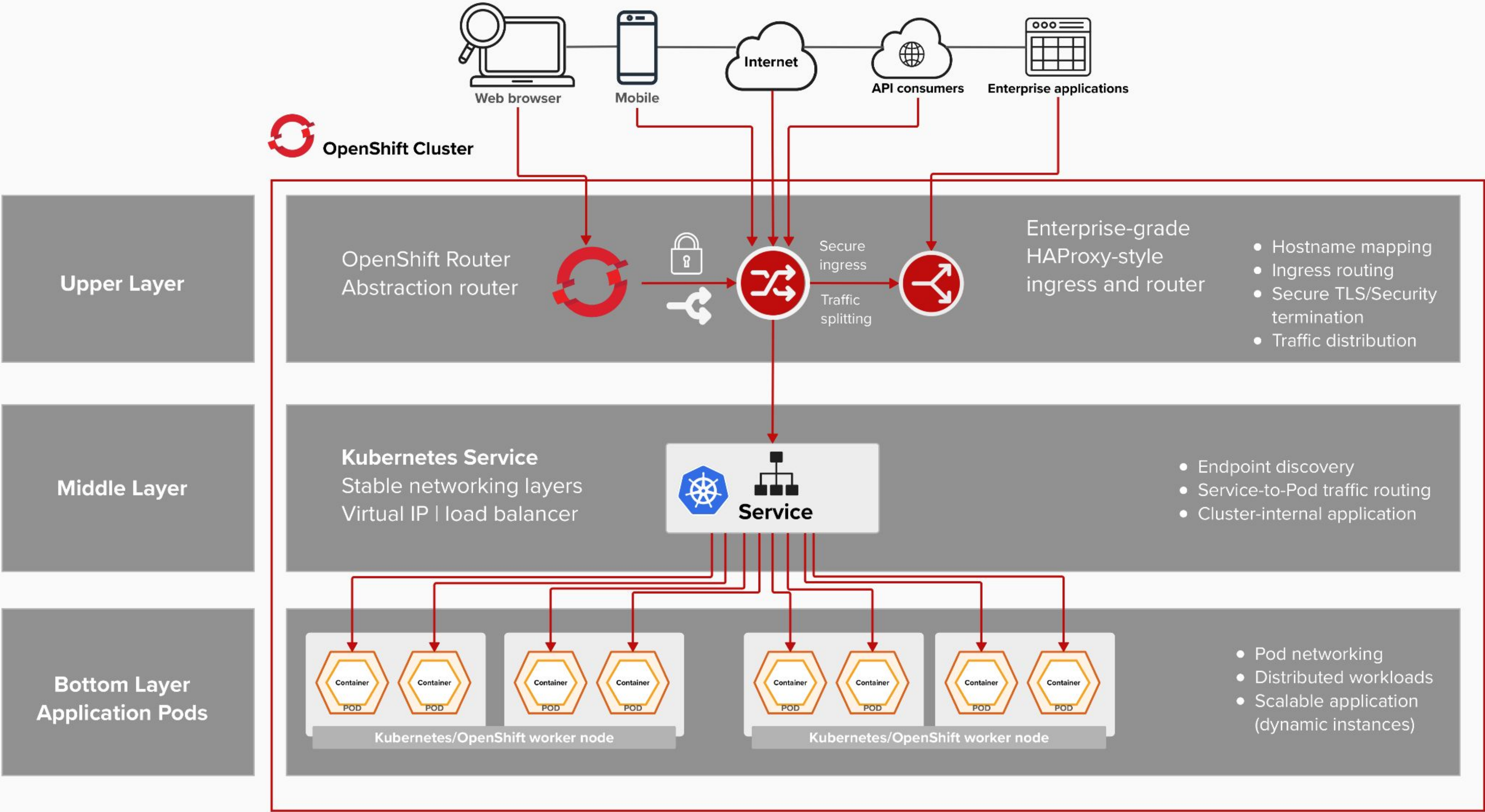
- RHEL CoreOS version & config
- Kubernetes core components
- OpenShift cluster components

Machine Config Operator (MCO) controls updates

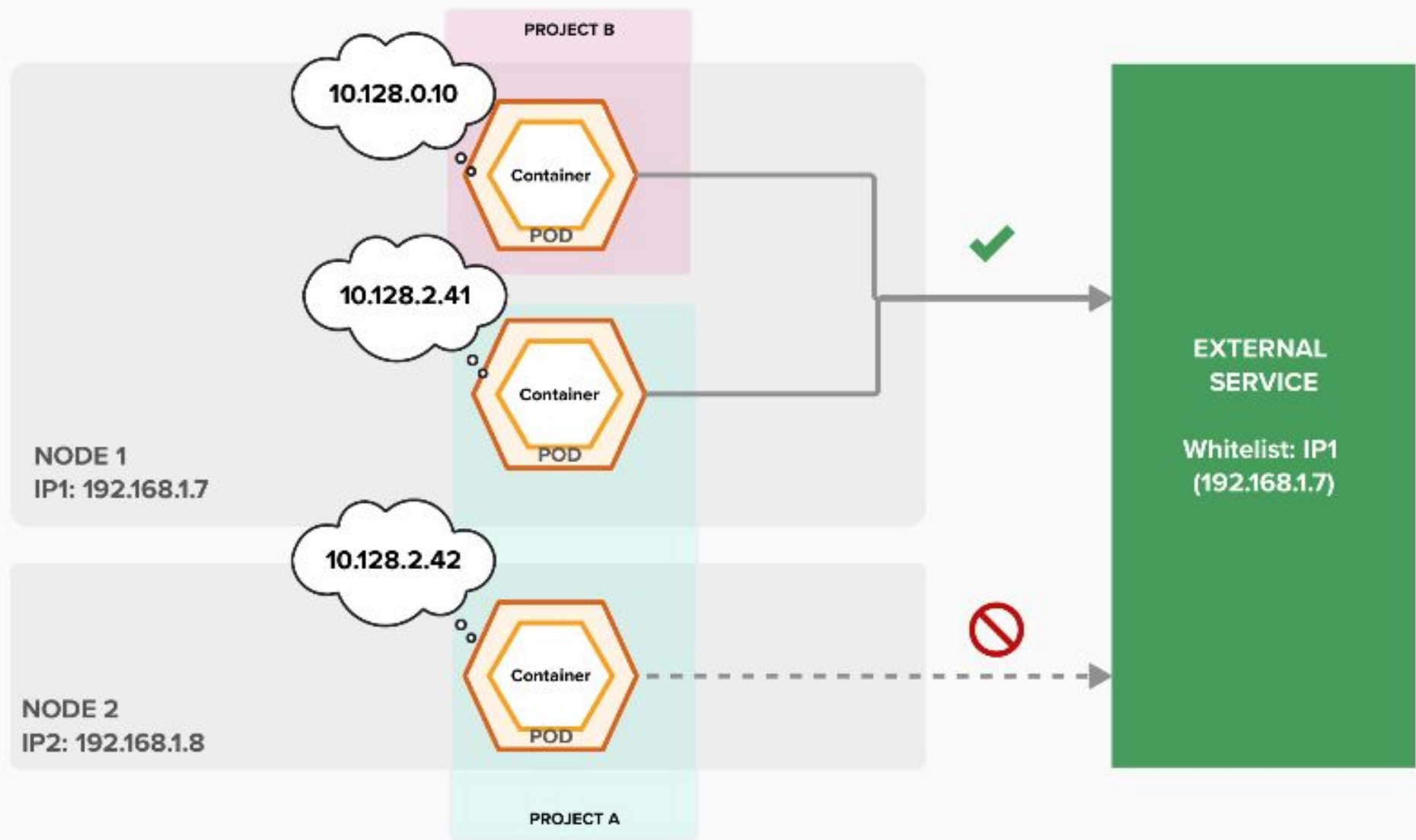
This is a DaemonSet that runs on all Nodes in the cluster. When you upgrade with `oc adm upgrade`, the MCO executes these changes.

Networking

Getting Traffic Into the Cluster



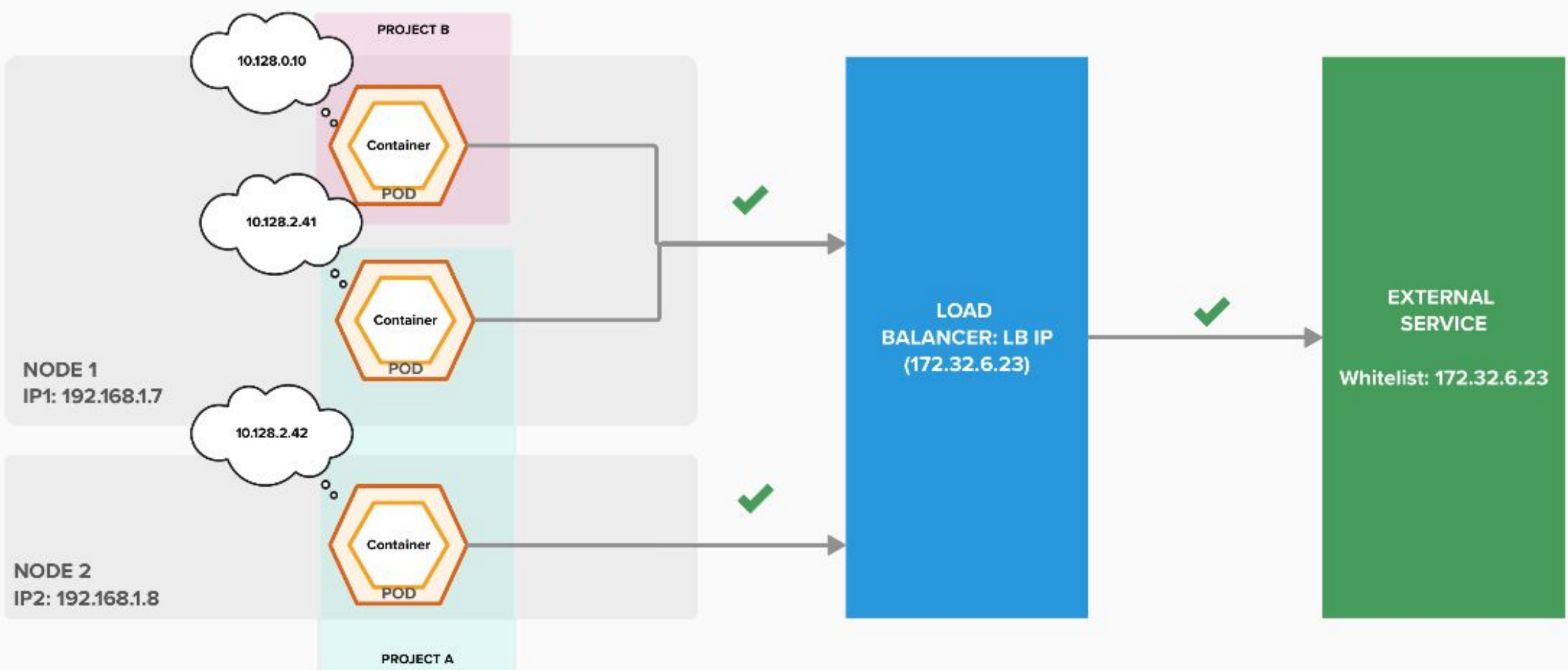
Getting Traffic Out the Cluster



SNAT

10.128.0.10 -> 192.168.1.7
10.128.2.41 -> 192.168.1.7

10.128.2.42 -> 192.168.1.8



SNAT

LB

10.128.0.10 -> 192.168.1.7 -> 172.32.6.23
10.128.2.41 -> 192.168.1.7 -> 172.32.6.23

10.128.2.42 -> 192.168.1.8 -> 172.32.6.23



Red Hat

Getting Traffic Around the Cluster

OpenShift Network Plugins

Subnet

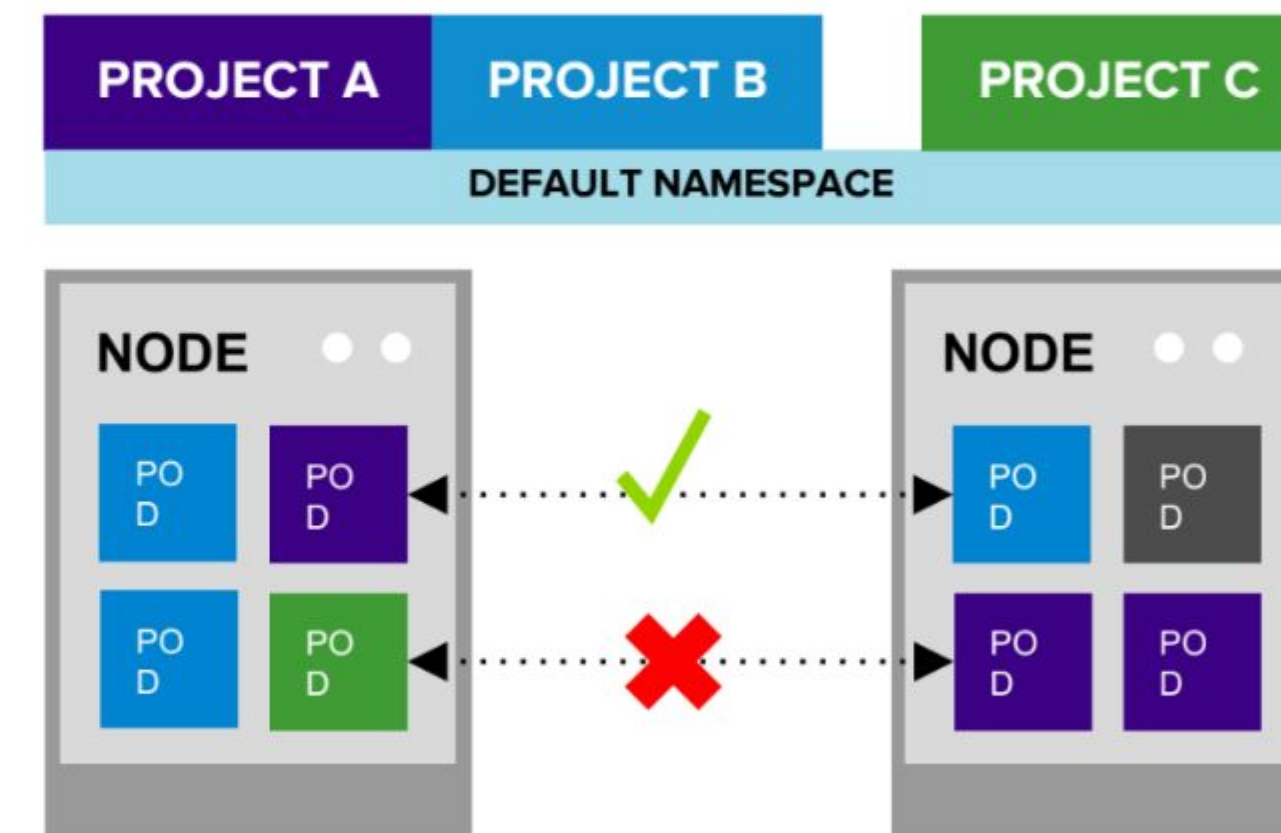
All pods can communicate with all other pods

Multitenant

Project level isolation.

Network Policy (Default)

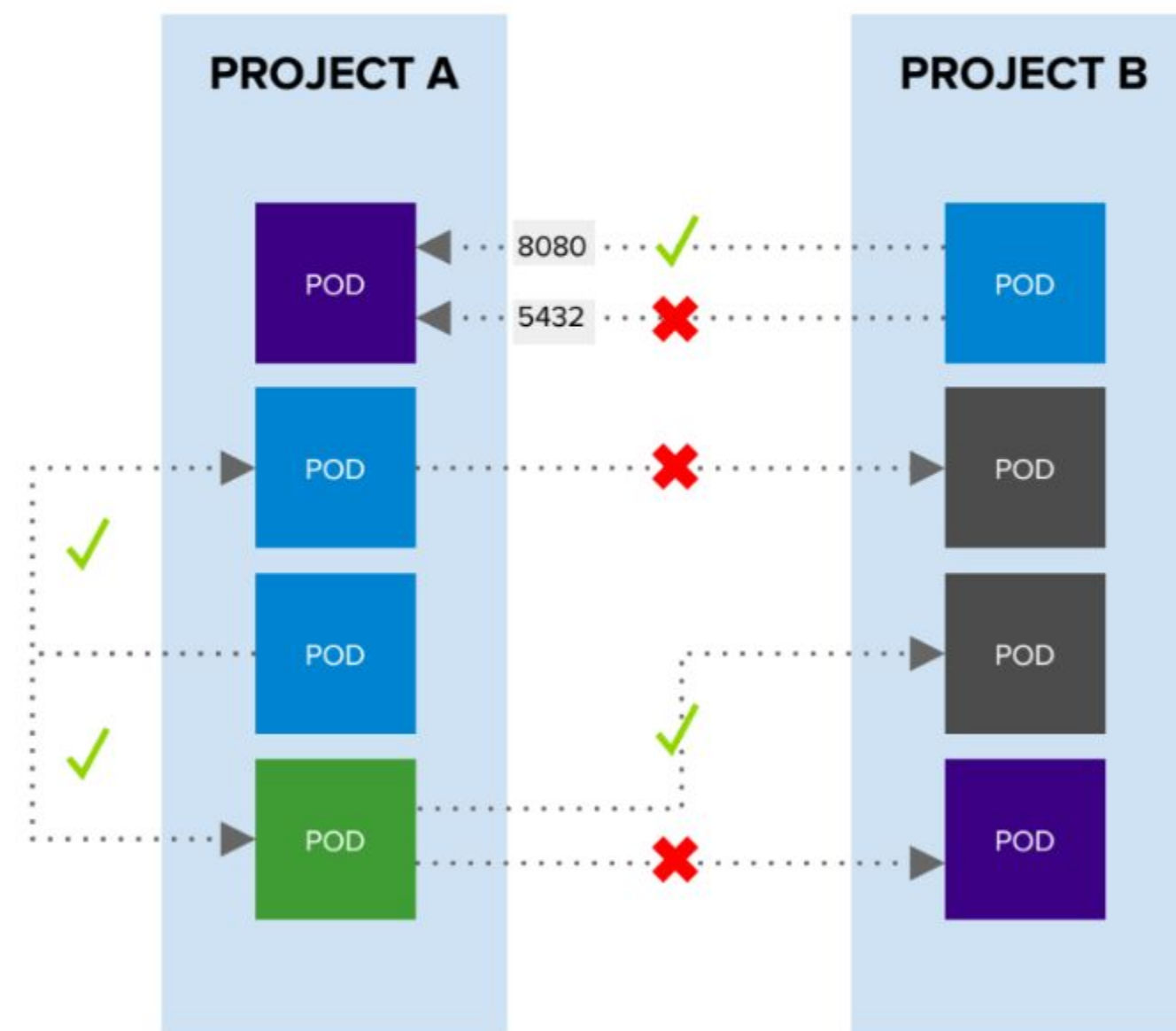
Granular policy based isolation



Multitenant Network

Getting Traffic Around the Cluster

Network Policy



Example Policies

Allow all traffic inside the project

Allow traffic from green to gray

Allow traffic to purple on 8080

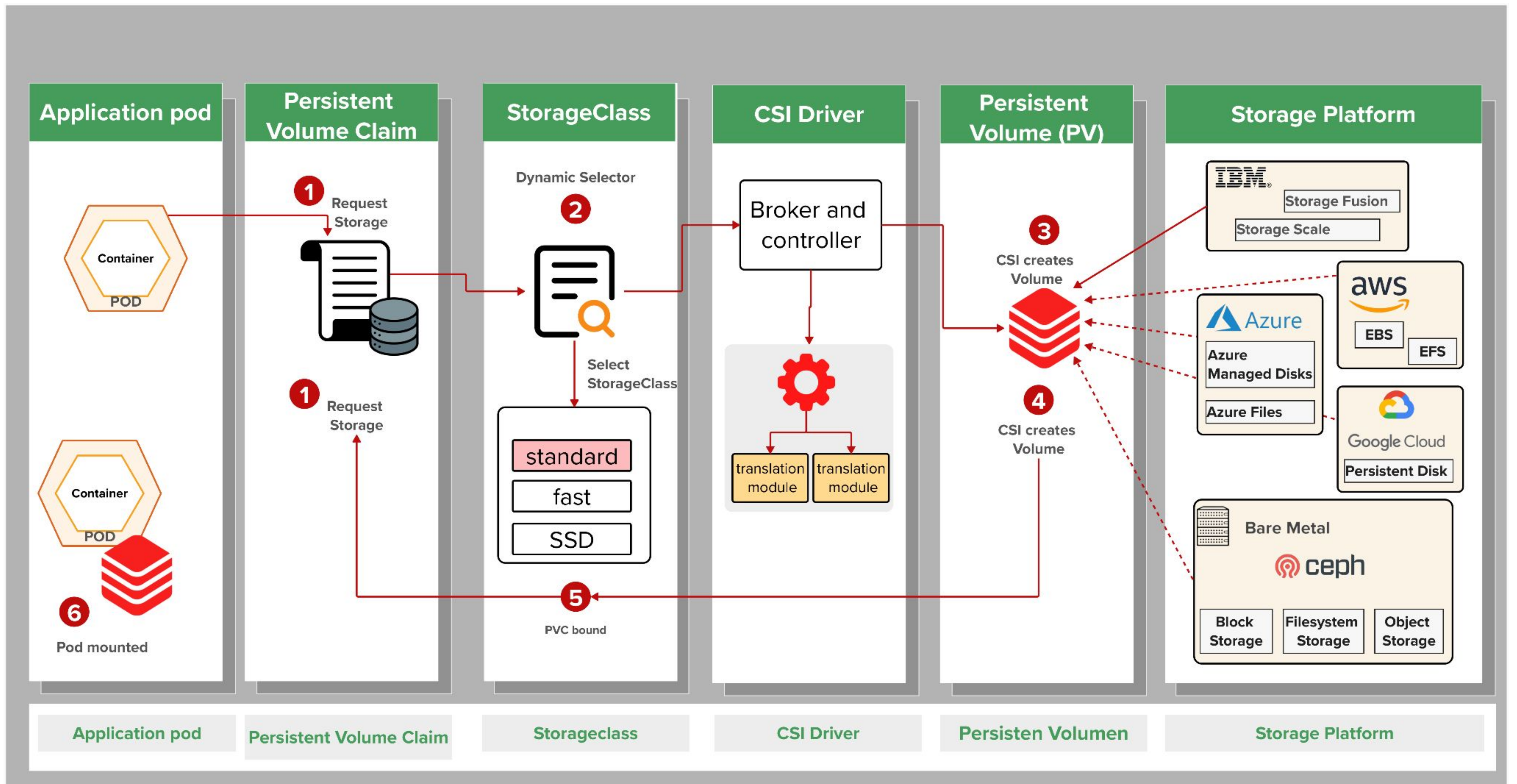
```
apiVersion: extensions/v1beta1
kind: NetworkPolicy
metadata:
  name: allow-to-purple-on-8080
spec:
  podSelector:
    matchLabels:
      color: purple
  ingress:
    - ports:
        - protocol: tcp
          port: 8080
```


OpenShift Security

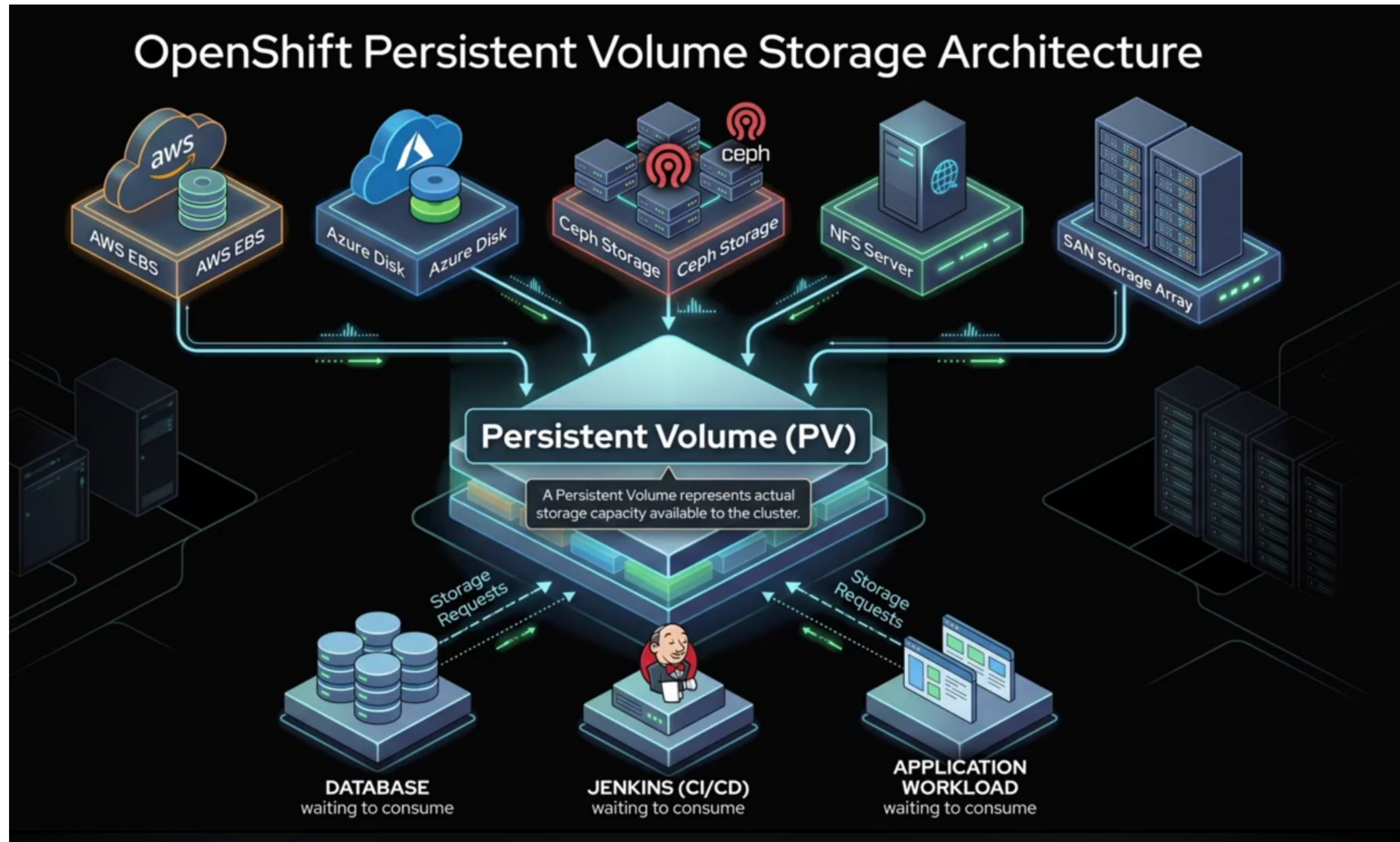
OpenShift Monitoring

OpenShift Logging

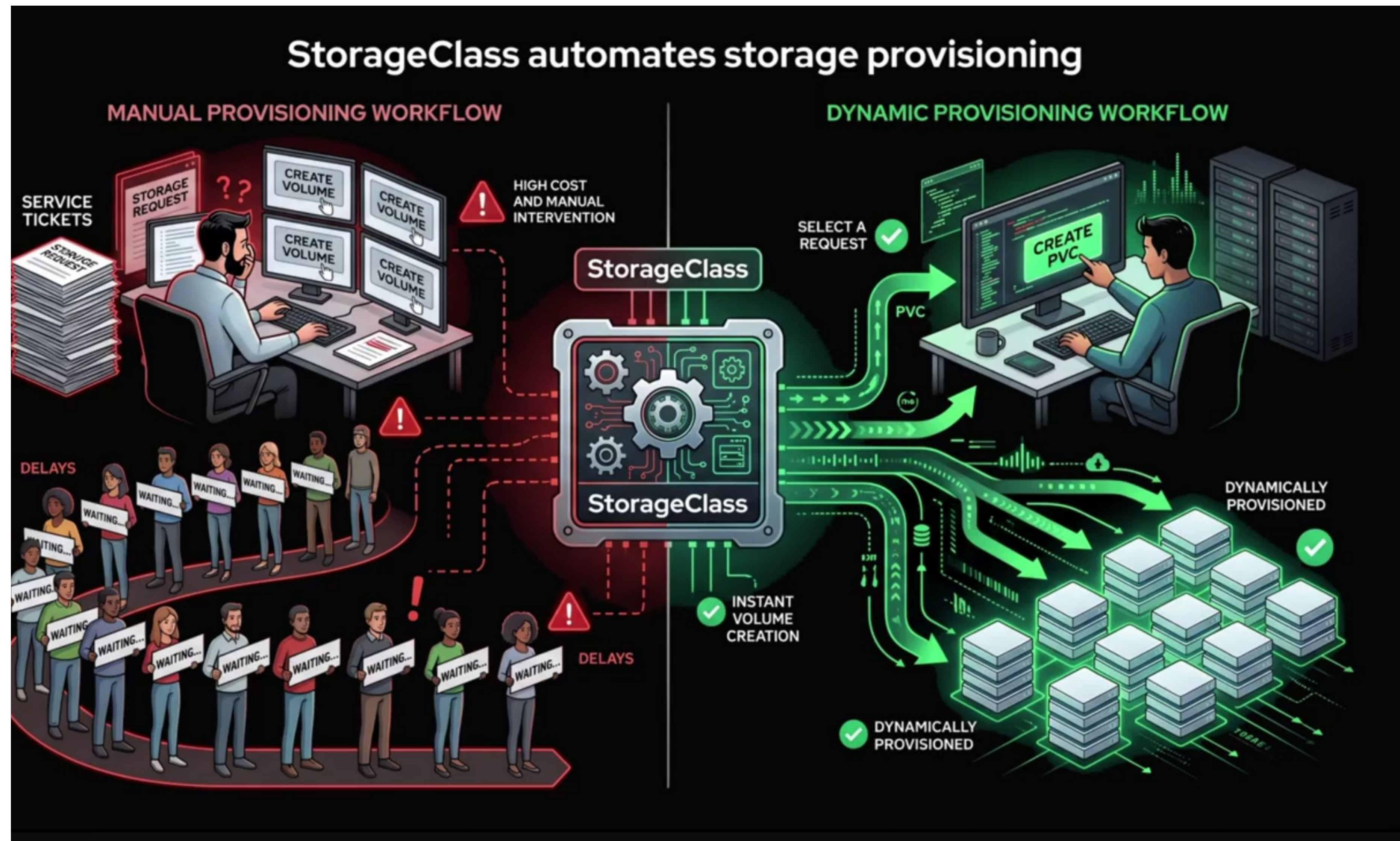
Persistent Storage



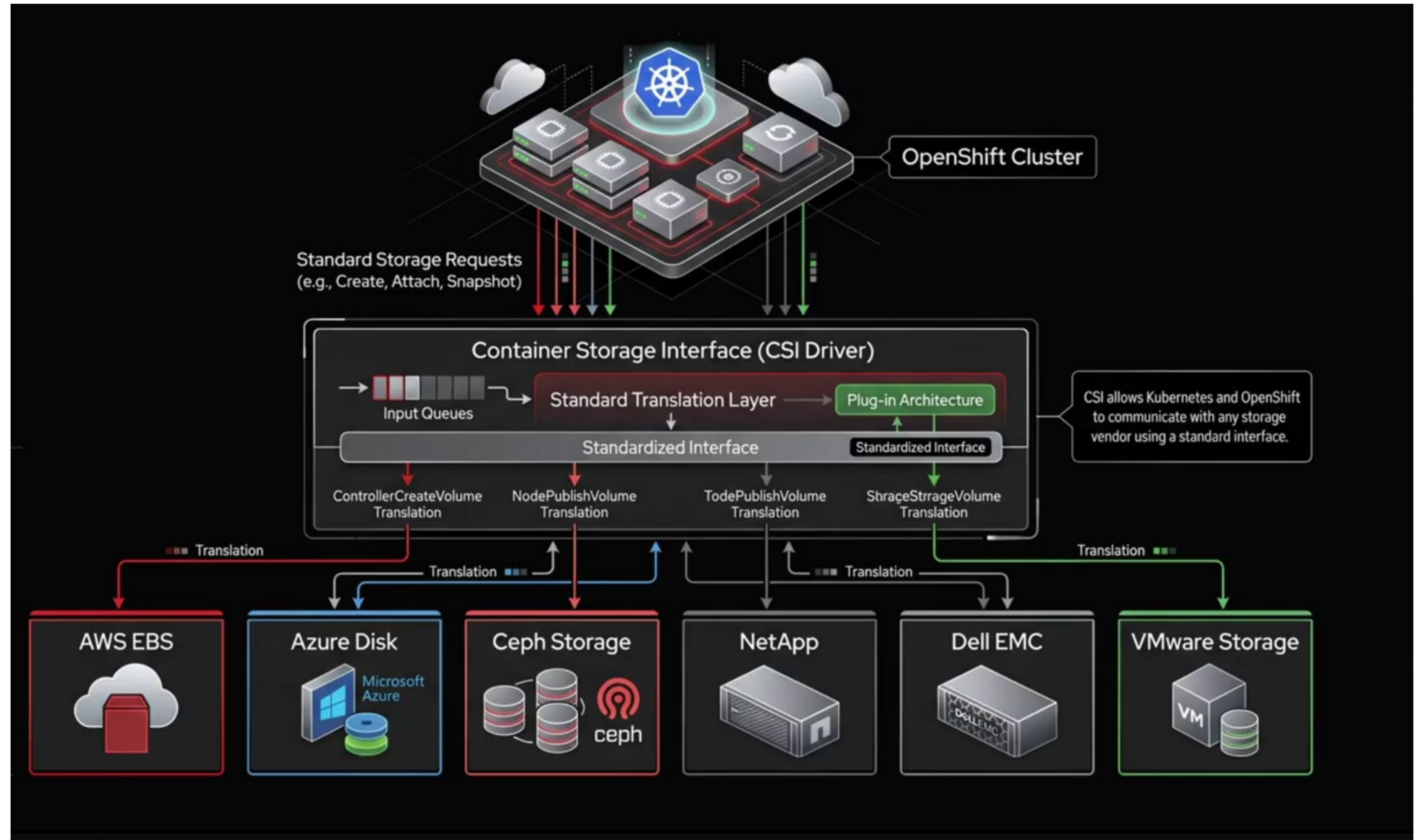
PV and PVC



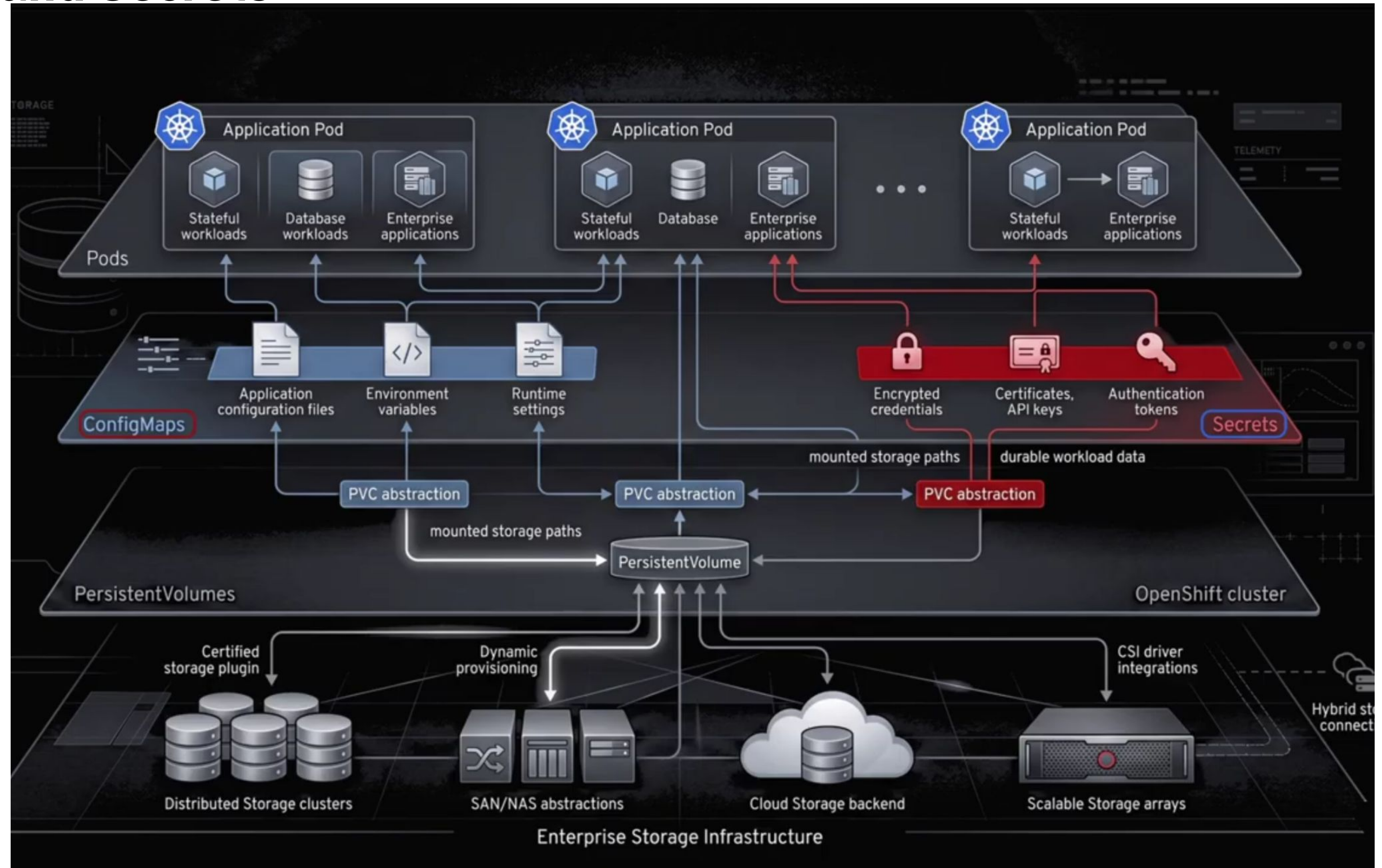
StorageClass



CSI Driver



ConfigMaps and Secrets



Operators

Red Hat Certified Operators

DEVOPS



APM



DATA SERVICES



DATABASE



SECURITY



STORAGE



Operator Catalog

All items

AI/Machine Learning

Application Runtime

Big Data

CI/CD

Cloud Provider

Database

Databases

Developer Tools

Drivers and plugins

Integration & Delivery

Languages

Logging & Tracing

Middleware

Modernization & Migration

Monitoring

Networking

Observability

OpenShift Optional

Security

Storage

Streaming & Messaging

All items

Filter by keyword...

Relevance

851 items

Helm Charts

.NET Application

Provided by Red Hat

A Helm chart to build and deploy .NET applications

Community

[DEPRECATED] Kuadrant Operator

Provided by Red Hat

[DEPRECATED] Follow <https://docs.kuadrant.io> instead

Community

3scale API Management

Provided by Red Hat

3scale Operator to provision 3scale and publish/manage API

Helm Charts

AAP Ansible automation portal

Provided by Red Hat

A Helm chart to deploy Ansible automation portal.

Red Hat

Advanced Cluster Management for Kubernetes

Provided by Red Hat

Advanced provisioning and management of OpenShift and Kubernetes clusters

Red Hat

Advanced Cluster Security for Kubernetes

Provided by Red Hat

Red Hat Advanced Cluster Security (RHACS) operator provisions the services necessar...

Red Hat

Ansible Automation Platform

Provided by Red Hat

The Ansible Automation Platform Resource Operator manages everything Automation

Templates

Apache HTTP Server

Provided by Red Hat, Inc.

An example Apache HTTP Server (httpd) application that serves static content. For more...

Community

APIcast

Provided by Red Hat

APIcast is an API gateway built on top of NGINX. It is part of the Red Hat 3scale API Management...

Community

Application Services Metering Operator

Provided by Red Hat

Collect the core usage of products from the Application Services portfolio into a single...

Operator Backed

Auth

Provided by Red Hat, Inc.

Auth is the Schema for the auths API

Operator Backed

AuthConfig

Provided by Red Hat

API to describe the desired protection for a service

Operator Backed

Authorino

Provided by Red Hat

API to create instances of authorino

Red Hat

Authorino Operator

Provided by Red Hat

Enables authentication and authorization for Gateways and applications in a Gateway API...

Community

Authorino Operator

Provided by Red Hat

The operator to manage instances of Authorino

Installed

Red Hat

AWS EFS CSI Driver Operator

Provided by Red Hat

Install and configure AWS EFS CSI driver.

Community

AWS EFS Operator

Provided by Red Hat

Manage read-write-many access to AWS EFS volumes in an OpenShift cluster.

Red Hat

AWS Load Balancer Operator

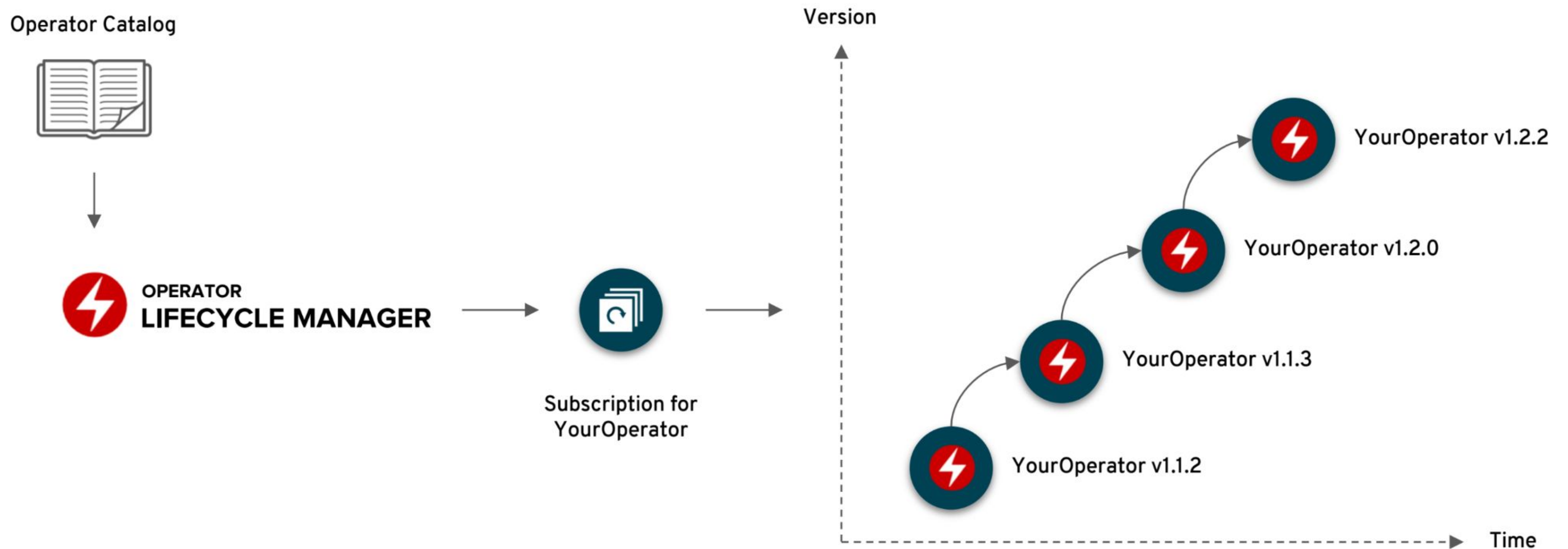
Provided by Red Hat Inc.

Operator to simplify the management of aws-load-balancer-controller. ###...

Operators as a First-Class Citizen

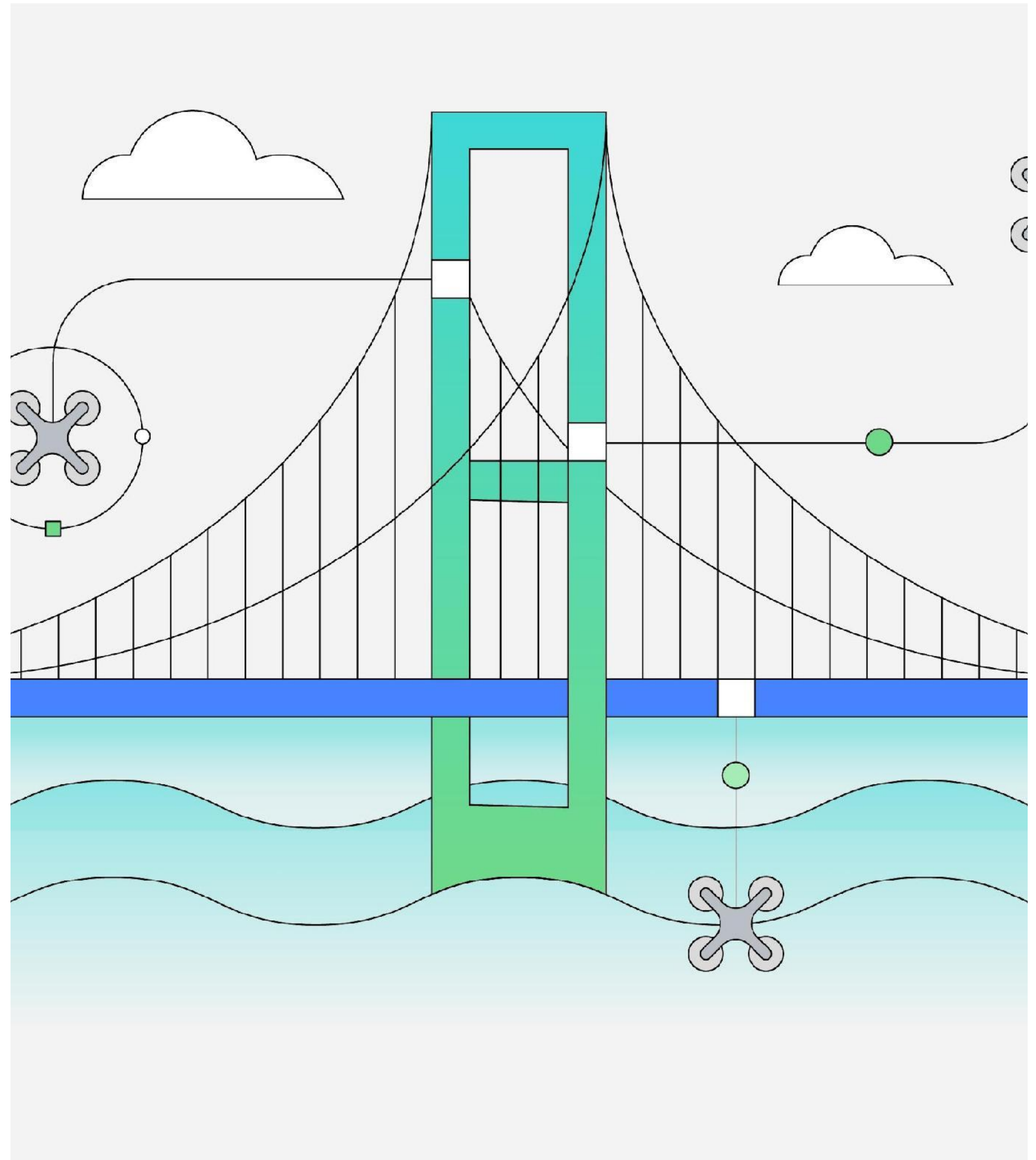
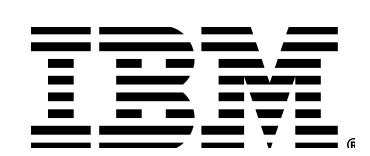


Operator Lifecycle Management



3.

text



4.

text

5.

text

text